HOLY CROSS COLLEGE

Life to the Full

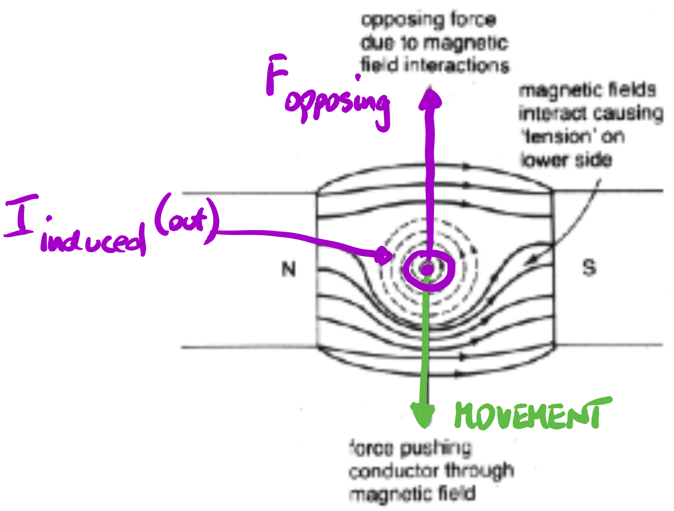
12 PHYSICS

INDUCED EMF & POWER TRANSMISSION

INDUCED EMF

Michael Faraday discovered that when a conductor is moved in a magnetic field, an electromotive force (EMF) is set up within it.
$\mathcal{E}$

Each of the free-moving electrons inside the metal has its own small magnetic field. This interacts with the external magnetic field (which is changing due to the movement of the conductor), causing a force to be exerted on them. Hence the electrons flow as a current in one direction.



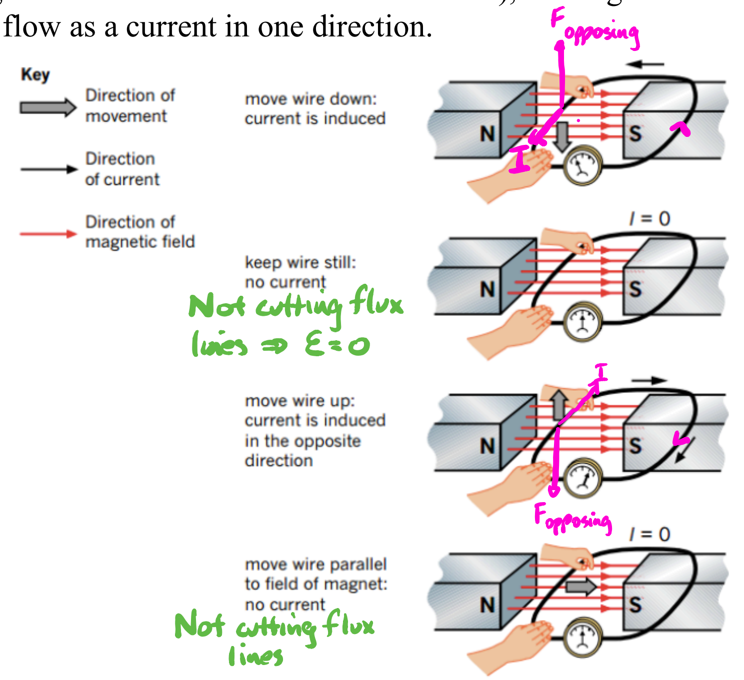


FIGURE 6.1.2 An electromotive force (emf) is induced in a wire when the wire moves perpendicular to a magnetic field.

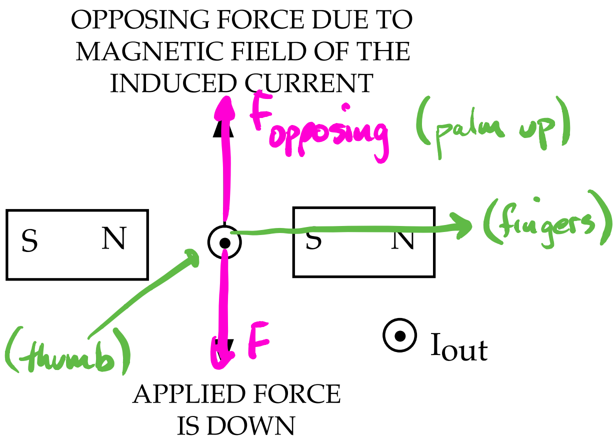
The conductor MUST cut flux lines to induce a current.

The key point is the changing magnetic field - if the magnetic field stops changing (i.e. the conductor stops moving), so too does the current stop.

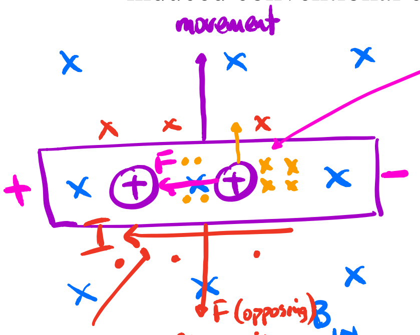
The induced current produces a magnetic field that interacts with the external field. The resulting force opposes the motion of the conductor.

Work has to be done to keep the conductor moving, and this work is seen as electrical energy in the conductor.

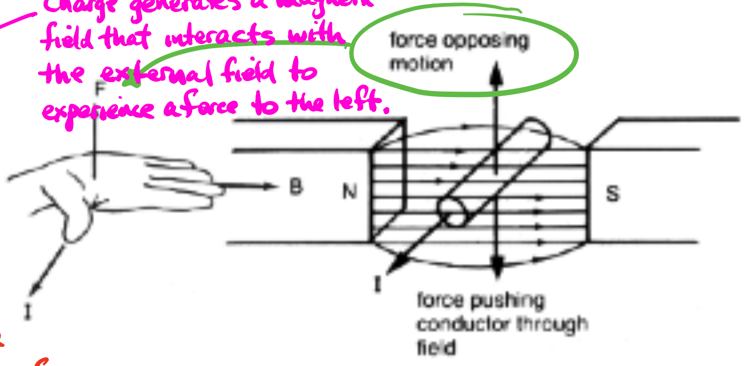
i.e. Mechanical work is transformed into electrical energy.



The direction of the induced current is given by the "right hand rule" - fingers in the direction of B; palm in the direction that ***opposes*** the motion of the conductor; thumb points in the direction of the induced conventional current I.



Charge generates a magnetic field that interacts with the external field to experience a force to the left.



When the current flows, it produces a magnetic field that interacts with the external field to produce an opposing force

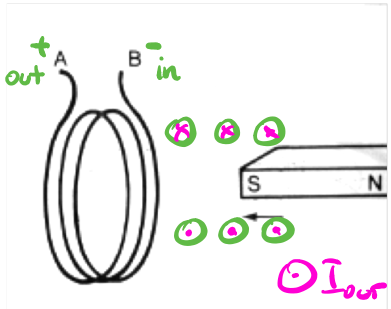
The production of an opposing force is summed up in Lenz's Law.

Lenz's Law: An induced current will be in such a direction as to produce a magnetic force that opposes the change causing it.

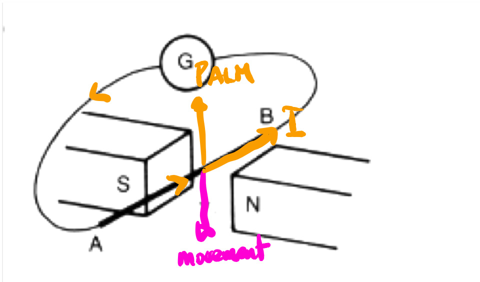
Examples

1. Apply Lenz's Law to each of the following situations.

(a)



(b)



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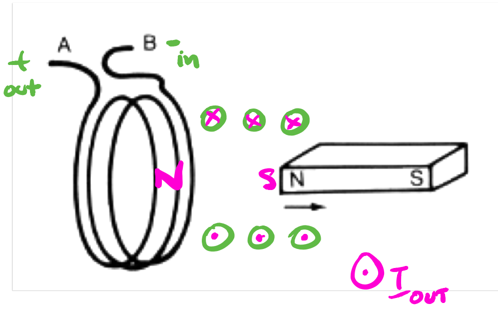

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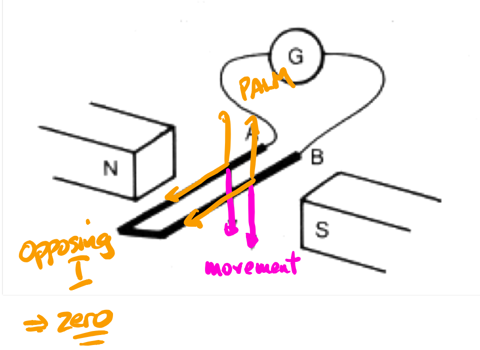
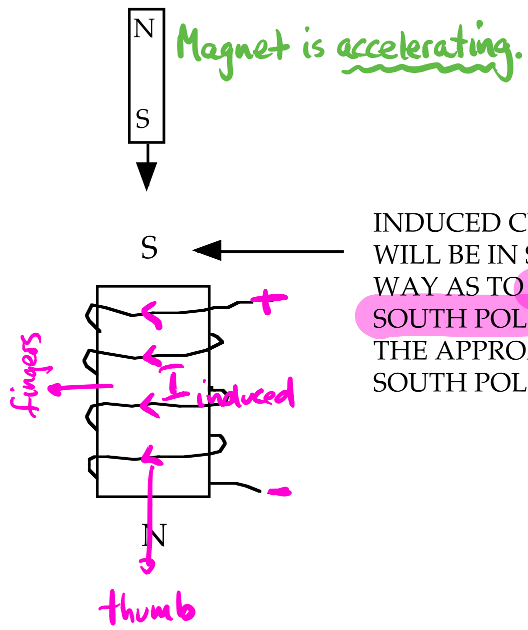
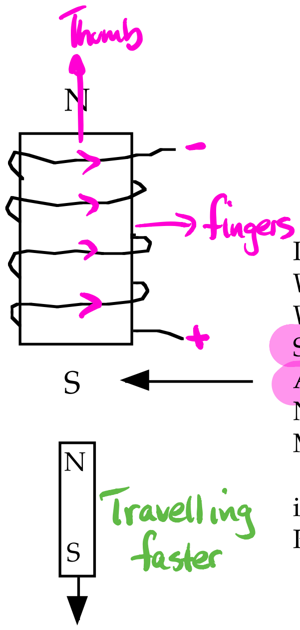


2. A magnet is dropped through the core of a solenoid as shown. Graph (without calculation) the induced current produced in the coil.

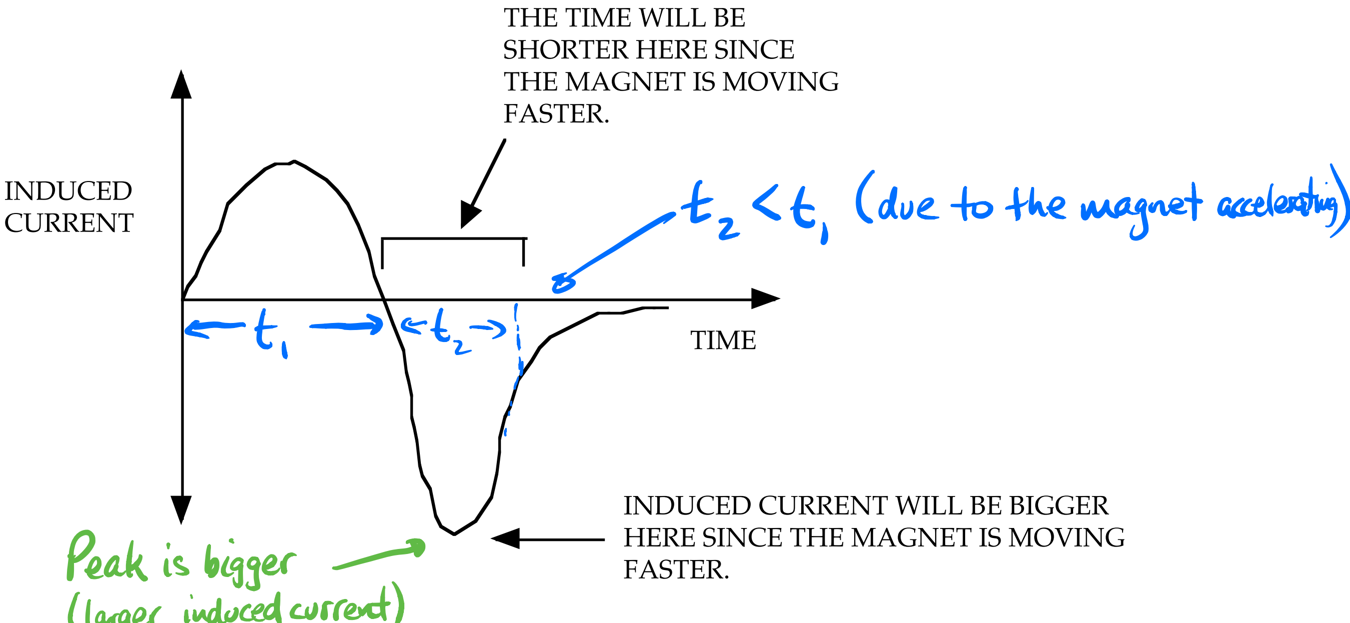


INDUCED CURRENT WILL BE IN SUCH A WAY AS TO PRODUCE A SOUTH POLE TO OPPOSE THE APPROACHING SOUTH POLE.



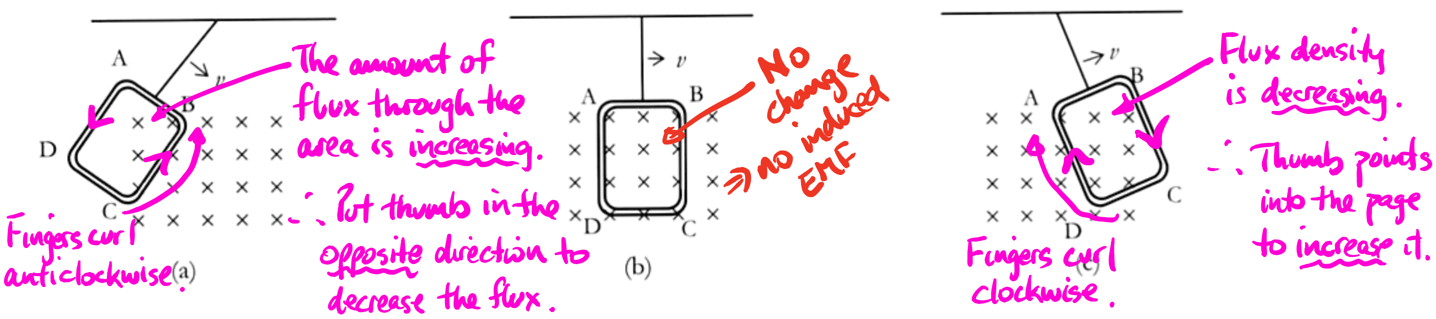
INDUCED CURRENT WILL BE IN SUCH A WAY AS TO PRODUCE A SOUTH POLE TO ATTRACT BACK THE NORTH POLE AS IT MOVES AWAY.

i.e. THE CURRENT REVERSES DIRECTION.



3. A rectangular copper coil ABCD is attached by a string to a support so that it can swing like a pendulum through a magnetic field as shown. The coil is initially released as shown in (a) and allowed to swing through the field. Draw in the direction of the induced currents.

Induced current is an eddy current.



INDUCED EMF IN A SINGLE CONDUCTOR

For a single conductor moving through a magnetic field, the induced EMF is given by:

$$\boxed{\textbf{EMF } (\varepsilon) = \textbf{Blv}}$$

where ε = the induced EMF across its ends (V).
B = the magnetic field strength (T).
l = the length of conductor cutting flux lines (m).
v = the velocity of the conductor through the field (ms^{-1}).

B, l and v are always at right angles to each other.

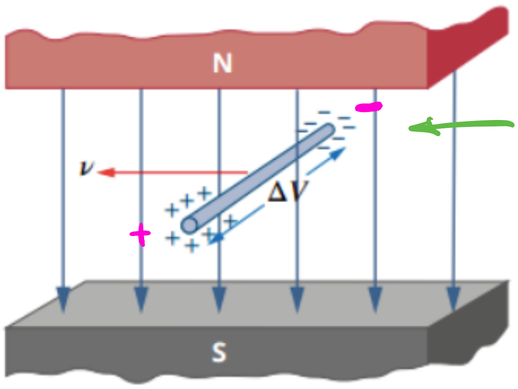
If any two of them are ***parallel*** to each other, then EMF = 0.

If any two are at an angle to each other (but not 90°), then a ***component*** will need to be used.

From this relationship, EMF can be increased by:

1. increasing the velocity of the conductor through the magnetic field.
2. increasing the magnetic field intensity.
3. increasing the length of wire cutting the magnetic flux lines.

If the ends of the conductor are connected in a circuit, the induced EMF will produce a current flow (though it will fluctuate with the speed and magnetic field intensity).



Get a potential difference across the ends.
No current flows unless it is a complete circuit.

FIGURE 6.1.3 A potential difference, ΔV, will be produced across a straight wire moving to the left in a downward-pointing magnetic field.

For a single conductor cutting a magnetic field at right angles, it sweeps an area given by:

$$\Delta A = l \times v\Delta t$$

Faraday's law

Area = $l \times s$

$= l \times vt$

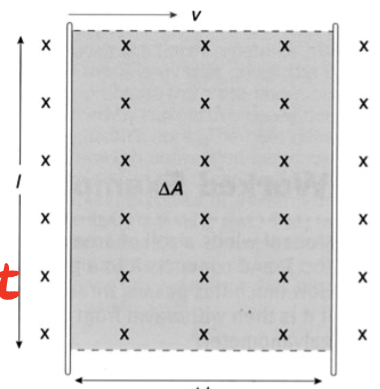
$$EMF = -N\frac{\Delta\Phi}{\Delta t} = -N\frac{B\Delta A}{\Delta t}$$

If $N = 1$, then:

$$EMF = \frac{Blv\Delta t}{\Delta t}$$

$\therefore$ <u>$EMF = Blv$</u>

No need to derive.

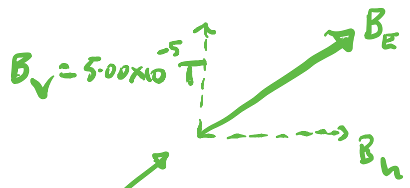


$s = vt$

Figure 4.8

The area ΔA (shaded) swept out by the wire in the field in a time Δt is equal to $lv\Delta t$.

Example 1



ELECTROMOTIVE FORCE ACROSS AN AIRCRAFT'S WINGS

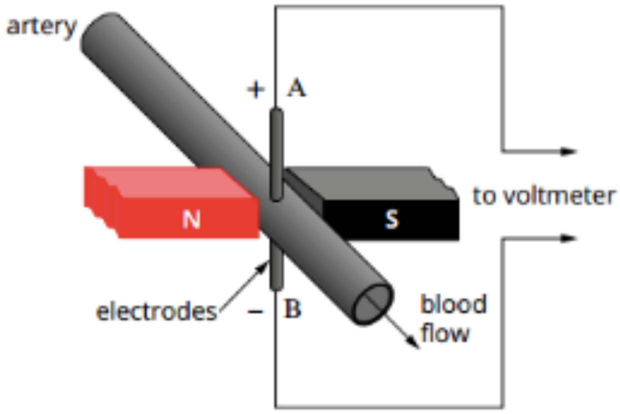
Will a moving aeroplane develop a dangerous emf between its wing tips solely from the Earth's magnetic field? An aircraft with a wingspan of 64.0 m is flying at a speed of 990 km h^{-1} at right angles to the Earth's magnetic field of 5.00×10^{-5} T.

Thinking	Working
Identify the quantities required in the correct units.	$\varepsilon = ?$ $\ell = 64.0$ m $B = 5.00 \times 10^{-5}$ T $v = 990$ km h^{-1} $= 990 \times \frac{1000}{3600} = 275$ m s^{-1}
Substitute into the appropriate formula and simplify.	$\varepsilon = \ell vB$ $= 64.0 \times 275 \times 5.00 \times 10^{-5} = 0.880$ V
State your answer as a response to the question.	$\varepsilon = 0.880$ V This is a very small emf and would not be dangerous.

Example 2

FLUID FLOW MEASUREMENT

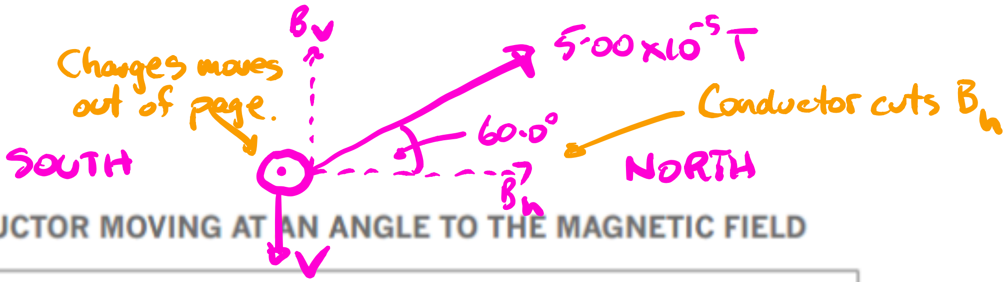
The rate of fluid flow within a vessel can be measured using the induced emf when the fluid contains charged ions. A small magnet and a sensitive voltmeter calibrated to measure speed are used. This can be applied to measure the flow of blood (which contains iron in solution) in the human body. If the diameter of a particular artery is 2.00 mm, the strength of the magnetic field applied is 0.100 T and the measured emf is 0.100 mV, what is the speed of the flow of the blood within the artery?



Convert

Thinking	**Working**
Identify the quantities required and put them into SI units.	$\varepsilon = 0.100\ \text{mV} = 1.00 \times 10^{-4}\ \text{V}$ $\ell = 2.00\ \text{mm} = 2.00 \times 10^{-3}\ \text{m}$ $v = ?$ $B = 0.100\ \text{T}$
Rearrange the appropriate formula, substitute and simplify.	$\varepsilon = \ell v B$ $v = \dfrac{\varepsilon}{\ell B}$ $= \dfrac{1.00 \times 10^{-4}}{2.00 \times 10^{-3} \times 0.100}$ $= 0.500\ \text{m s}^{-1}$
State your answer with the correct units.	$v = 0.500\ \text{m s}^{-1}$ The speed of the flow of the blood within the artery is $0.500\ \text{m s}^{-1}$.

Example 3



EMF ACROSS CONDUCTOR MOVING AT AN ANGLE TO THE MAGNETIC FIELD

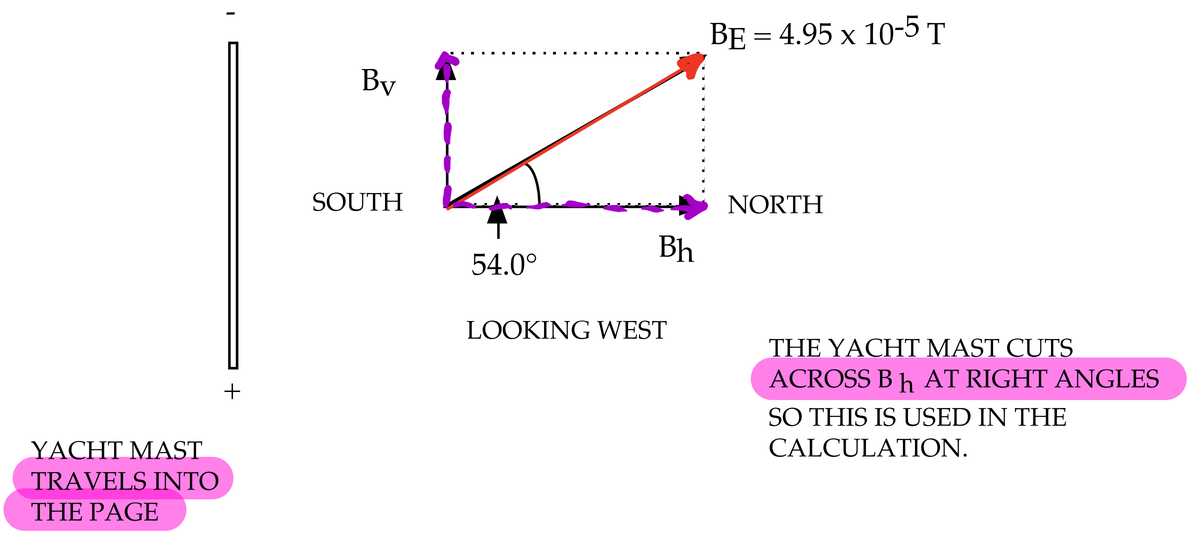
A horizontal conducting rod of length 10.0 cm and facing east–west is dropped vertically where the Earth's magnetic field is 5.00×10^{-5} T at an angle of dip of 60.0°. Calculate the induced emf between its ends when it reaches a velocity of 20.0 m s^{-1}.	
Thinking	**Working**
Identify the component of magnetic field that will cause an emf to be induced.	If the rod is falling vertically, it will not cut the vertical component of magnetic field. The horizontal component of the magnetic field needs to be found.
Calculate the horizontal component of the magnetic field.	$B_h = B\cos\theta$ $= 5.00 \times 10^{-5}\cos 60°$ $= 2.50 \times 10^{-5}$ T
Identify the quantities required and put them into SI units.	$\varepsilon = ?$ $\ell = 10.0$ cm $= 0.100$ m $v = 20.0$ m s^{-1} $B_h = 2.50 \times 10^{-5}$ T
Use the appropriate formula, substitute and calculate answer.	$\varepsilon = \ell v B$ $= 0.100 \times 20.0 \times 2.50 \times 10^{-5}$ $= 5.00 \times 10^{-5}$
State your answer with the correct units.	$\varepsilon = 5.00 \times 10^{-5}$ V

$$\text{EMF} = B_h \ell v$$
$$= (5{\cdot}00 \times 10^{-5} \cos 60{\cdot}0°)(0{\cdot}100)(20{\cdot}0)$$
$$= \underline{5{\cdot}00 \times 10^{-5}\ \text{V}}$$

Example 4

A yacht with a 6.00 m high mast is sailing at 4.00 ms^{-1} due west in the Southern Hemisphere where the Earth's magnetic field intensity is 4.95 x 10^{-5} T at 54.0° dip.

(a) Calculate the EMF induced across the ends of the mast.
(b) Determine which end of the mast is the positive end.



$$\begin{aligned} \text{EMF} &= Blv \\ &= (4.95 \times 10^{-5} \cos 54.0°)(6.00)(4.00) \\ &= 6.983 \times 10^{-4} \text{ V} \end{aligned}$$

The induced EMF = 6.98 x 10^{-4} V.

(b) Using the right hand rule (with the palm coming **out** of the page to oppose the motion and the fingers pointing north in the direction of B_h), the thumb points down.

Since this indicates the direction of positive charge flow, the positive charges move down to this end while the negative charges move to the top. This causes a charge separation with the **bottom end being positive**.

Example 5

As an aeroplane flies along through the Earth's magnetic field, it is acting rather like the straight wire described above.

a At which places on the Earth will the induced EMF across an aeroplane's wings be maximum?

b What is the maximum EMF which would be induced across the wings of a Boeing 747 with a wing span of 64.0 m, flying at 990 km h^{-1} through the Earth's magnetic field? Take the maximum magnitude of the Earth's magnetic field as 60.0 μT.

Solution

a As the plane flies horizontally, it cuts through the vertical component of the Earth's field. Near the magnetic poles, the field is close to vertical, so this is where we would expect the maximum induced EMF. Near the Equator the Earth's field is nearly horizontal and so the plane is flying parallel to the field lines and does not cut them; thus there will be no induced EMF along the wing.

b The maximum magnitude of the induced EMF near the poles is found from:

$$v = \frac{990}{3.6}\ \text{m s}^{-1} \qquad \text{EMF} = v\boldsymbol{B}_{\perp}l$$

$$\boldsymbol{B}_{\perp} = 60.0 \times 10^{-6}\ \text{T} \qquad = (275)(60.0 \times 10^{-6})(64.0)$$

$$l = 64.0\ \text{m} \qquad = 1.06\ \text{V}$$

Not only is this EMF very small, but any attempt to use it, or even measure it, would be thwarted by the fact that the same EMF will be induced in any wires connecting the wing tips, giving us two equal but opposite EMFs in the circuit.

Law of Induction

Faraday demonstrated that an induce EMF is produced when ***magnetic flux lines are cut per unit time***.

> ***Faraday's Law of Induction***: The size of the induced EMF depends on the rate of change of magnetic flux linked with the coil.

$$\boxed{\text{EMF } (\varepsilon) = \frac{-N\Delta\Phi}{\Delta t}}$$

$\frac{\Delta\phi}{\Delta t}$ = flux linkage

where
- ε = the induced EMF (V).
- N = the number of loops in the coil.
- $\Delta\Phi$ = the change in flux (Wb).
- Δt = the time in which the change takes place (s).

Note

The negative sign shows the relative direction of the induced EMF and is related to the laws of energy conservation.

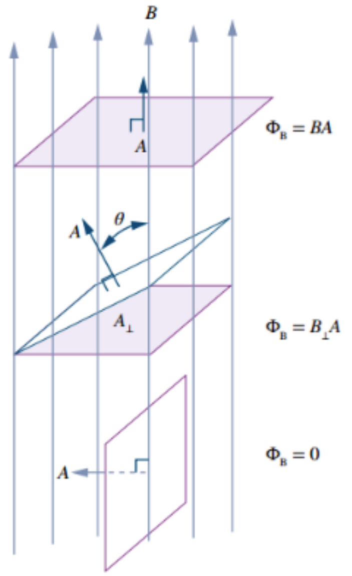


FIGURE 6.2.3 The magnetic flux is the strength of the magnetic field, B, multiplied by the area perpendicular to the magnetic field, given by $A\cos\theta$ and shown as the shaded areas in the diagrams above.

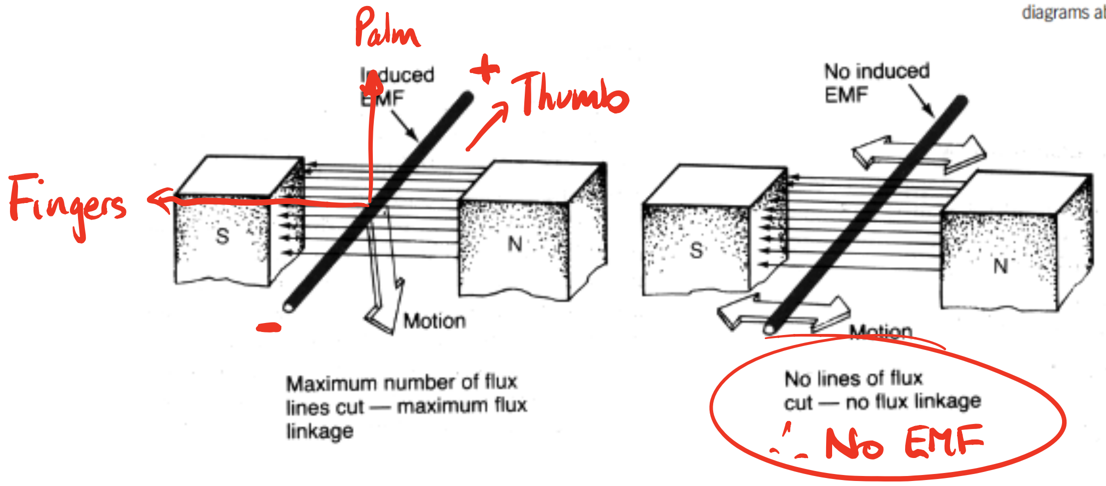
$\Delta\Phi/\Delta t$ is called the ***flux linkage*** or rate of change of magnetic flux. It is a measure of how many flux lines are cut by the conductor per unit time.

An induced EMF is produced by:

- a wire or coil moving through a magnetic field. (The more loops in the coil, the greater the induced EMF.)
- moving a magnet in relation to a stationary wire or coil.
- varying the magnetic field strength to alter $\Delta\Phi/\Delta t$.

No induced current is produced when:

- the conductor is stationary.
- the conductor moves ***parallel*** to the magnetic field.



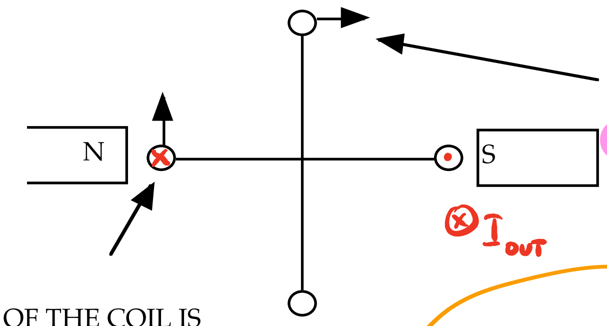
Examples

1. **(Electrical Generator)** Convert

A coil of dimensions 20.0 cm x 10.0 cm and containing 200 loops (N) is pivoted along its long axis in a horizontal magnetic field of 0.342 T (B). It is rotated at 30.0 r.p.m. Calculate the maximum EMF generated by the coil.

$T = \frac{60\,s}{30.0} = 2.00\,s$

$f = \frac{30.0}{60} = 0.500\,Hz$



IN THIS POSITION, THE SIDE OF THE COIL IS MOVING PARALLEL TO THE MAGNETIC FIELD AND NOT CUTTING ANY FLUX LINES.

=> **NO EMF**

THE SIDE OF THE COIL IS CUTTING ACROSS THE MAXIMUM NUMBER OF FLUX LINES WHEN THE COIL IS IN THIS POSITION.

=> **MAXIMUM EMF**

THE INDUCED EMF DROPS FROM MAXIMUM TO ZERO IN 1/4 OF A TURN.

=> **FIND THE TIME FOR 1/4 TURN** WHEN DOING THIS SORT OF CALCULATION.

Calculates EMF (average)

To calculate the period for one revolution:

$$T = \frac{60.0}{30.0} = 2.00\text{ s}$$

$$\Rightarrow \quad \Delta t = 1/4\,(2.00) = 0.500\text{ s}$$

Show this in the final calculation.

To calculate $\Delta\Phi$, consider the number of flux lines that thread through the area of the coil as it moves through 1/4 of a turn - it goes from maximum to zero.

$$\Rightarrow \quad \Delta\Phi = B\Delta A = (0.342)(0.200)(0.100) = 6.84 \times 10^{-3}\text{ Wb}$$

Show in final calculation.

Area changes while B is constant

$$\text{EMF} = \frac{-N\,\Delta\Phi}{\Delta t} = \frac{-(200)(6.84 \times 10^{-3})}{(0.500)} = -2.736\text{ V}$$

The induced EMF = 2.74 V .

$$\text{EMF} = \frac{-N\Delta\Phi}{\Delta t} = \frac{-NB\Delta A}{\Delta t} = -\frac{(200)(0.342)[0-(0.200)(0.100)]}{\frac{1}{4}(2.00)} = 2.74\text{ V}$$

Note

This equation gives the ***average EMF*** generated in the coil, since the edge of the coil moves from cutting the maximum number of flux lines (max EMF) to cutting no flux lines (zero EMF).

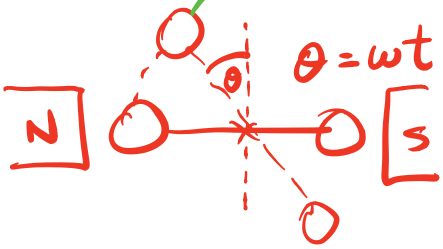
relates to frequency.

Following is the derivation of the equation to give the ***maximum or instantaneous EMF*** in a rotating coil, taking into account its position relative to the magnetic field.

Consider the rotation speed of the coil. It is represented by the angular speed **ω**, which is measured in radians s^{-1}.

ω (angular speed)

i.e. If the ***angle θ is the angle of the coil to the vertical***:



$\theta = \omega t$

$$\theta = \omega t$$

The amount of flux threading through a coil is given by:

$$\phi = BA\cos\theta$$
$$= \underline{BA\cos\omega t}$$

Using Calculus, the EMF induced due to the rate of change of the flux through the coil is given by:

$EMF = \dfrac{-N\Delta\phi}{\Delta t}$

$$EMF = \frac{-Nd\phi}{dt}$$
$$= \frac{-Nd(BA\cos\omega t)}{dt}$$
$$\Rightarrow EMF = NBA\omega\sin\omega t \quad ✓$$

One complete revolution can be thought of as:

$$\theta = 360° = 2\pi \text{ radians}$$

The time taken for one revolution is the period T, and the frequency of the rotation is given by:

$$f = \frac{1}{T}.$$

$$\therefore \quad \omega = \frac{2\pi}{T} = 2\pi f$$

Must use frequency.

$$\therefore \quad \boxed{\textbf{EMF }(\varepsilon) = \textbf{NBA}2\pi\textbf{f}\sin\theta}$$

On data sheet

where θ = *the angle of the coil to the vertical*.

2. **Microphone and Guitar String**

The function of a microphone is to convert sound vibrations into electrical signals.

The so-called 'dynamic' microphone uses a tiny coil attached to a diaphragm, which vibrates with the sound.

The coil vibrates within the magnetic field of a permanent magnet, thus producing an induced EMF that varies with the original sound.

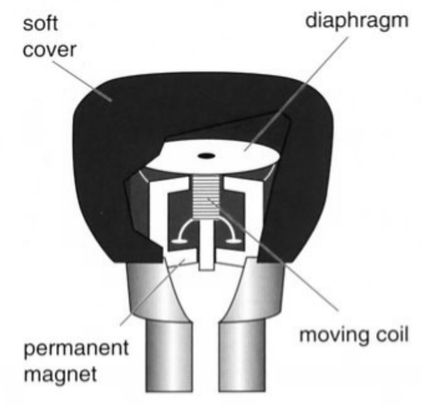


Figure 4.1
Sound waves vibrate the diaphragm of a microphone. These vibrations are converted to a tiny electric current by the relative motion of the coil and permanent magnet.

Has been asked in WACE paper.

This field interacts with the coil.

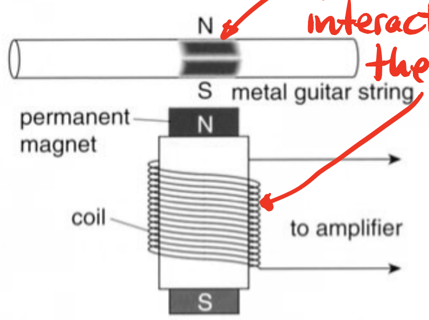


Figure 4.2
A guitar pick-up magnetises the steel string with a permanent magnet and then uses this vibrating magnet to induce an EMF in a coil below it.

The electric guitar uses electromagnetic induction to generate an electric signal which is then amplified electronically.

The pick-up consists of a coil around a small permanent magnet. The permanent magnet induces the section of the wire above it to become a magnet.

As the string vibrates, this moving magnet induces an EMF in the coil. This is then amplified, and often electronically 'doctored', to produce the sound required.

3. **Stationary coil - changing field.**

Changing B

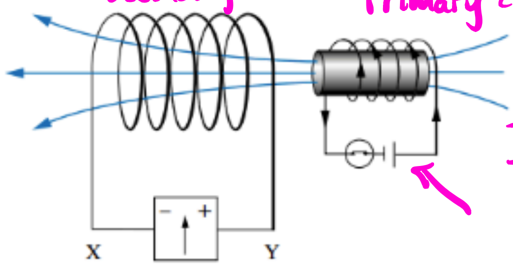
$\therefore EMF = -\frac{N\Delta\phi}{\Delta t} = -\frac{N\Delta BA}{\Delta t}$

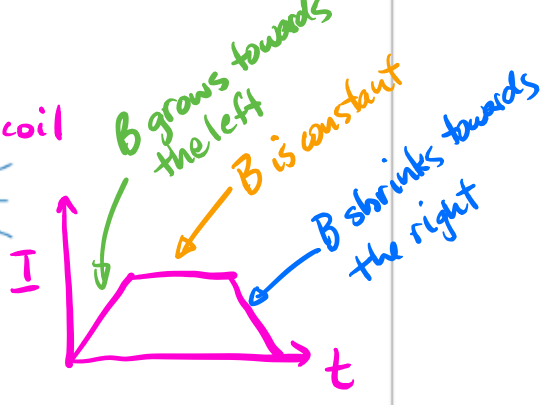
INDUCED CURRENT IN A COIL FROM AN ELECTROMAGNET

Instead of using a permanent magnet to change the flux in the loop in Worked example 6.3.1, an electromagnet (on the right, in the diagram below) could be used. What is the direction of the current induced in the solenoid when the electromagnet is:

(i) switched on?
(ii) left on?
(iii) switched off?

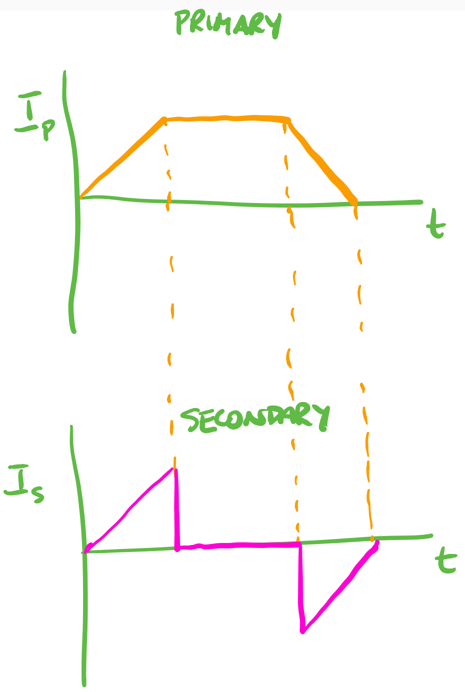
What happens?





Thinking	Working
Consider the direction of the change in magnetic flux for each case.	**(i)** Initially there is no magnetic flux through the solenoid. When the electromagnet is switched on, it creates a magnetic field directed to the left. So the change in flux through the solenoid is increasing to the left. **(ii)** While the current in the electromagnet is steady, the magnetic flux through the solenoid is constant and the flux is not changing. **(iii)** In this case, initially there is a magnetic flux through the solenoid from the electromagnet directed to the left. When the electromagnet is switched off, there is no longer a magnetic flux through the solenoid. So the change in flux through the solenoid is decreasing to the left.
What will oppose the change in flux for each case?	**(i)** The magnetic field that opposes the change in flux through the solenoid is directed to the right. **(ii)** There is no change in flux and so there will be no opposition needed and no magnetic field created by the solenoid. **(iii)** The magnetic field that opposes the change in flux through the solenoid is directed to the left.
Determine the direction of the induced current required to oppose the change for each case.	**(i)** In order to oppose the change, the current will flow through the solenoid in the direction from X to Y (or through the meter from Y to X), using the right-hand grip rule. **(ii)** There will be no induced emf or current in the solenoid. **(iii)** In order to oppose the change, the current will flow through the solenoid in the direction from Y to X (or through the meter from X to Y), using the right-hand grip rule.

$EMF = -\frac{N\Delta\phi}{\Delta t} = \frac{-N\Delta BA}{\Delta t}$



4. **Coil pulled out of a field.**

A student winds a coil of area 40.0 cm² with 20 turns. He places it horizontally in a vertical uniform magnetic field of 0.100 T, and connects it to a galvanometer with resistance 200.0 Ω.

a How much flux passes through the coil?

b If it is then withdrawn from the field in a time of 0.500 s, what would be the average current reading on the galvanometer?

Solution

a $B_\perp = 0.100$ T $\quad\quad \Phi_B = B_\perp A$

$A = \dfrac{40.0}{100 \times 100}$ m² $\quad\quad = (0.100)(4.00 \times 10^{-3})$

$= 4.00 \times 10^{-4}$ Wb

b The average EMF induced in each turn is given by the average rate of change of flux:

$\Delta\Phi_B = 4.00 \times 10^{-4}$ Wb $\quad\quad \text{EMF} = -N\dfrac{\Delta\Phi_B}{\Delta t}$

$\Delta t = 0.500$ s $\quad\quad = -(20)\dfrac{4.00 \times 10^{-4}}{0.500}$

$N = 20$ turns $\quad\quad = -1.60 \times 10^{-2}$ V

Assuming that the only significant resistance in the circuit is the galvanometer, the average induced current is given by Ohm's law:

$\Delta V = 1.60 \times 10^{-2}$ V $\quad\quad I = \dfrac{\Delta V}{R}$

$R = 200.0\ \Omega$ $\quad\quad = \dfrac{1.60 \times 10^{-2}}{200.0}$

$= 8.00 \times 10^{-5}$ A

5. **Coil in a field.**

A student places a horizontal 5.00 cm square coil of wire into a uniform vertical magnetic field of 0.100 T.

a How much flux cuts through the coil?

She then pulls the coil out of the field at such a rate that it takes 1.00 s for the coil to lose all the flux. While the coil is coming out of the field, she finds that a current of +2.00 mA is registered on the ammeter connected to it.

b If she puts the coil back into the flux at the same rate, what current will she observe?

c If she pulls it out again, but twice as fast as before, what current will be induced?

Now she steadily rotates the loop while it is fully in the field in such a way that it takes 2.00 s to rotate through 180.0°.

d Describe the current that will be induced.

Solution

a $B = 0.100$ T $\quad\quad \Phi_B = BA_\perp = (0.100)(0.0500^2)$

$A_\perp = 0.0500^2$ m² $\quad\quad = 2.50 \times 10^{-4}$ Wb

b The same current but in the opposite direction: −2.00 mA.

c The induced current is related to the change in flux over the change in time so:

$$I_{ind} \propto \frac{\Delta\Phi_B}{\Delta t}$$

This time Δt was halved so the current will be doubled. The current will be +4.00 mA.

d The total change of flux that occurs in the 2.00 s is:

$$\Delta\Phi_B = \Phi_{B\,\text{final}} - \Phi_{B\,\text{initial}}$$
$$= (-2.50 \times 10^{-4}) - (2.50 \times 10^{-4})$$
$$= -5.00 \times 10^{-4}\ \text{Wb}$$

As the flux changes from $+2.50 \times 10^{-4}$ to -2.50×10^{-4} Wb, the sign of the flux is related to the side of the coil it passes through. As the time is doubled the average rate of this flux change is the same as in part a ($-2.50 \times 10^{-}$ Wb s^{-1}), so the average current will again be ±2.00 mA. However, this will not be uniform. It will be a maximum (±2.00 mA) as the loop passes through the vertical and a minimum as the loop is horizontal. The magnitude of the current is not related to the magnitude of the flux, but to the magnitude of the rate of change of flux.

6.

NUMBER OF TURNS IN A COIL

A coil of cross-sectional area $1.00 \times 10^{-3}\,m^2$ experiences a change in the strength of a magnetic field from 0.00 to 0.200 T over 0.500 s. If the magnitude of the average induced emf is measured as 0.100 V, how many turns must be on the coil?	
Thinking	**Working**
Identify the quantities needed to calculate the magnetic flux through the coil when in the presence of the magnetic field and convert to SI units where required.	$\Phi = BA_{\perp}$ $B = 0.200\,T$ $A = 1.00 \times 10^{-3}\,m^2$
Calculate the magnetic flux when in the presence of the magnetic field.	$\Phi = BA_{\perp}$ $= 0.200 \times 1.00 \times 10^{-3}$ $= 2.00 \times 10^{-4}\,Wb$
Identify the quantities from the question required to complete Faraday's law.	$\varepsilon = -N\frac{\Delta\Phi}{t} = \frac{-N\Delta BA}{\Delta t}$ $N = ?$ $\Delta\Phi = \Phi_2 - \Phi_1$ $= 2.00 \times 10^{-4} - 0$ $= 2.00 \times 10^{-4}\,Wb$ $\Delta t = 0.500\,s$ $\varepsilon = 0.100\,V$ $\Rightarrow N = -\frac{\varepsilon \Delta T}{\Delta BA}$ $= \frac{-(0.100)(0.500)}{(0-0.200)(1.00\times10^{-3})}$ $= 250$ turns
Rearrange Faraday's law and solve for the number of turns on the coil. Ignore the negative sign.	$\varepsilon = -N\frac{\Delta\Phi}{t}$ $N = -\frac{\varepsilon t}{\Delta\Phi}$ $= \frac{0.100 \times 0.500}{2.00 \times 10^{-4}}$ $= 250$ turns

7. **Induced current by changing area.**

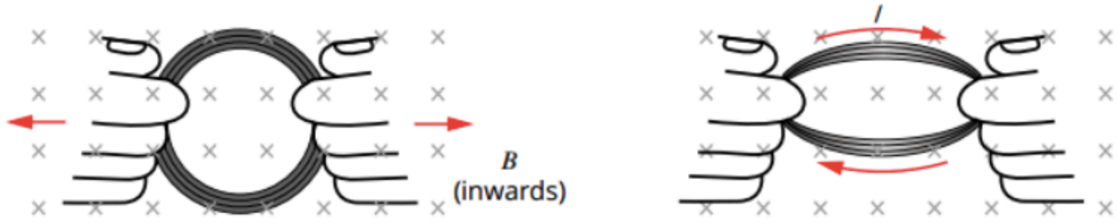


FIGURE 6.3.4 Inducing a current by changing the area of a coil. The amount of flux (the number of field lines) through the coil is reduced and an emf is therefore induced during the time that the change is taking place. The current flows in a direction that creates a field to oppose the reduction in flux into the page.

$$EMF = -N\frac{\Delta\phi}{\Delta t} = -NB\frac{\Delta A}{\Delta t}$$

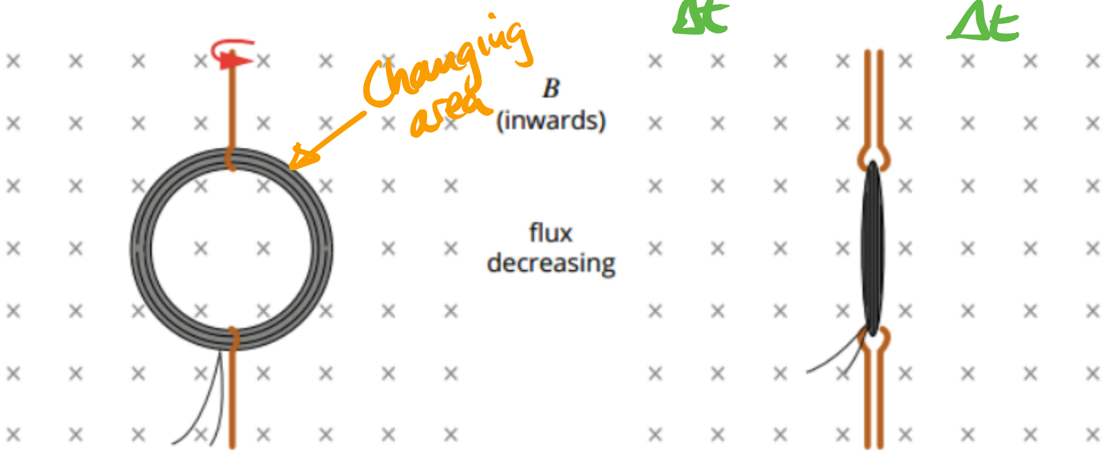


FIGURE 6.3.5 Changing the orientation of a coil within a magnetic field by rotating it reduces the amount of flux through the coil and so induces an emf in the coil while it is being rotated.

Applications

EXTENSION

Superconductors and superconducting magnets

Technological breakthroughs have often led to advances in physics. This was the case in 1908 when Kamerlingh Onnes, at the University of Leiden in the Netherlands, succeeded in liquefying helium. Helium liquefies at 4.2 K (–268.9°C). It was known that the electrical resistance of metals decreases as they cool, so one of the first things that Onnes and his assistant did was to measure the resistance of some metals at these very low temperatures.

Onnes was hoping to find that as the temperature of mercury dropped towards absolute zero its resistance would also gradually drop towards zero. What they found, however, was a complete surprise. At 4.2 K its resistance vanished completely!

Onnes coined the word 'superconductivity' to describe this phenomenon. Soon he found that some other metals also became superconducting at extremely low temperatures: lead at 7.2 K and tin at 3.7 K, for example. Curiously, metals such as copper and gold, which are very good conductors at normal temperatures, do not become superconducting at all. Onnes was awarded the 1913 Nobel Prize in Physics for his work in low-temperature physics.

Much of the great promise of superconductivity has to do with the magnetic properties of superconductors. In a superconductor an induced current does not fade away. As the resultant field opposes the changing flux, the magnet is repelled. This gives rise to the 'magnetic levitation' effect that is by now well known (Figure 6.3.8). On a large scale this could perhaps one day provide a virtually frictionless maglev (magnetic levitation) train.

Unfortunately, the early superconducting metals lost their superconductivity in moderate magnetic fields of about 0.1 T or less. However, in the 1940s it was found that some alloys of elements such as niobium had higher 'critical temperatures' and, more particularly, retained their properties in stronger magnetic fields. By 1973 the niobium–germanium alloy Nb_3Ge held the record with a critical temperature of 23.2 K in a magnetic field of 38 T, an extremely strong field.

In 1986 an entirely new and exciting class of superconductors was discovered. Georg Bednorz and Karl Müller, working in Switzerland, found that compounds of some rare earth elements and copper oxide had considerably higher critical temperatures. They received the 1987 Nobel Prize in Physics for their work.

These new 'warm superconductors' are ceramic materials made by powdering and baking the metal compounds. Most ceramics are insulators; it was a combination of good science and inspired guesswork that led Müller to try such unlikely candidates for superconductivity. So far, the record is held by bismuth and thallium oxides with a critical temperature around 125 K—still rather cold, but significantly above the temperature of readily available liquid nitrogen (77 K).

Superconductivity, particularly in the newer materials, is still not fully understood. It can really only be discussed in terms of quantum physics, but one rather picturesque way of thinking about it is that electrons pair up and 'surf' electrical waves set up by each other in the crystal lattice of the material.

The promise of superconductivity is enormous: low friction transport, no-loss transmission of electricity, and smaller and more powerful electric motors and generators. Superconducting magnets might be used to contain the extremely hot plasma needed to bring about hydrogen fusion, producing almost pollution-free energy in much the same way that the Sun does. Superconducting magnets of between 2 T and 10 T are regularly used in medical applications such as magnetic resonance imaging and in research.

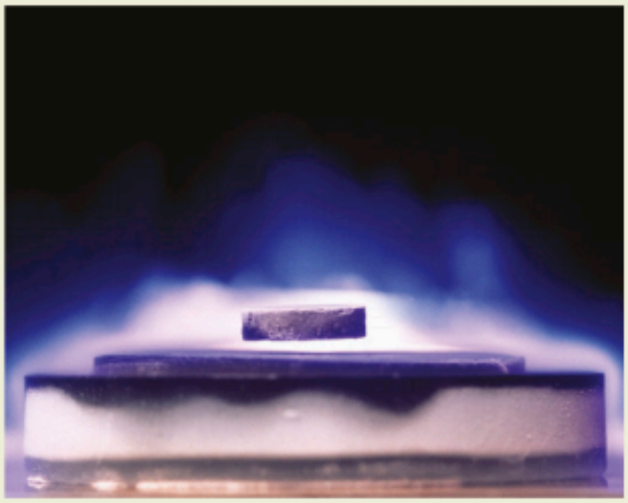
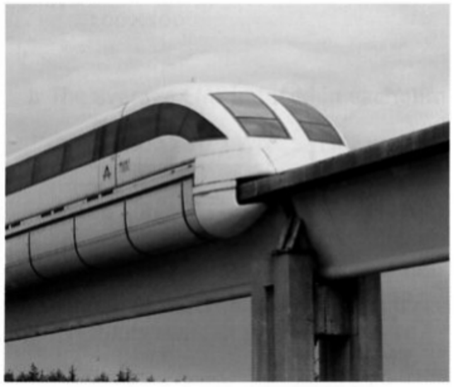


FIGURE 6.3.8 A disc magnet is repelled by a superconductor because the magnet induces a permanent current in the superconductor, which results in an opposing field.

Maglev Trains



A maglev train is supported by magnetic forces created by induced, or eddy, currents. Various systems are being trialled in several countries. Generally the force is between currents induced in flat conductors under the train by strong, possibly superconducting, magnets mounted in the fast-moving train—normal suspension is needed for slow travel. There are two components to the force: one is an undesirable drag force we normally notice when a conductor is moved slowly near a magnet; the other is the repulsive lift force. Fortunately, the drag force decreases with speed and the lift force increases with speed. At around 300 km h^{-1} the lift force is about 50 times stronger than the drag force.

Induction Cooker

PHYSICS IN ACTION

Induction stoves

In contrast to conventional gas or electric stoves, which heat via radiant heat from a hot source, an **induction hotplate** heats via an eddy current in the metal pot in which the food is being cooked. A coil of copper wire is placed within the cooktop (Figure 6.3.7). The AC electricity supply produces a changing magnetic field in the coil. This induces an eddy current in the conductive metal pot. The resistance of the metal in the pot, in which the eddy current flows, transforms electrical energy into heat and cooks the food.

Although induction cooktops have only reached the domestic market in relatively recent times, the first patents for induction cookers where issued in the early 1900s. They have significant advantages over traditional electric cooktops in that they allow instant control of cooking power (similar to gas burners), they lose less energy through ambient heat loss and heating time, and they have a lower risk of burn injuries. Overall, the heating efficiency of an induction cooktop is about 12% better than traditional electric cooktops and twice that of gas.

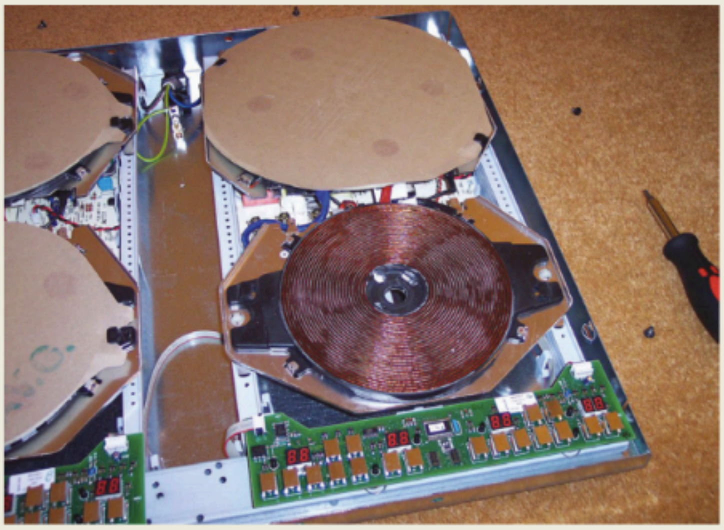
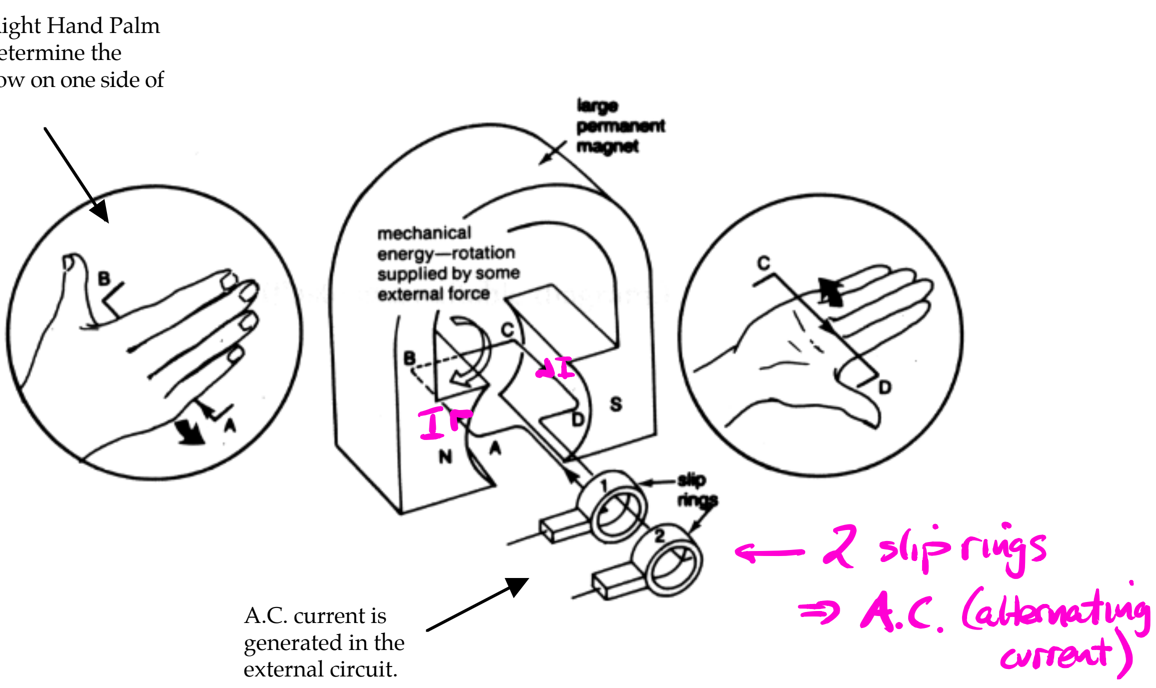


FIGURE 6.3.7 The coil of an induction zone within an induction cooktop. The large copper coil creates an alternating magnetic field.

THE GENERATOR

Alternating Current (A.C.)

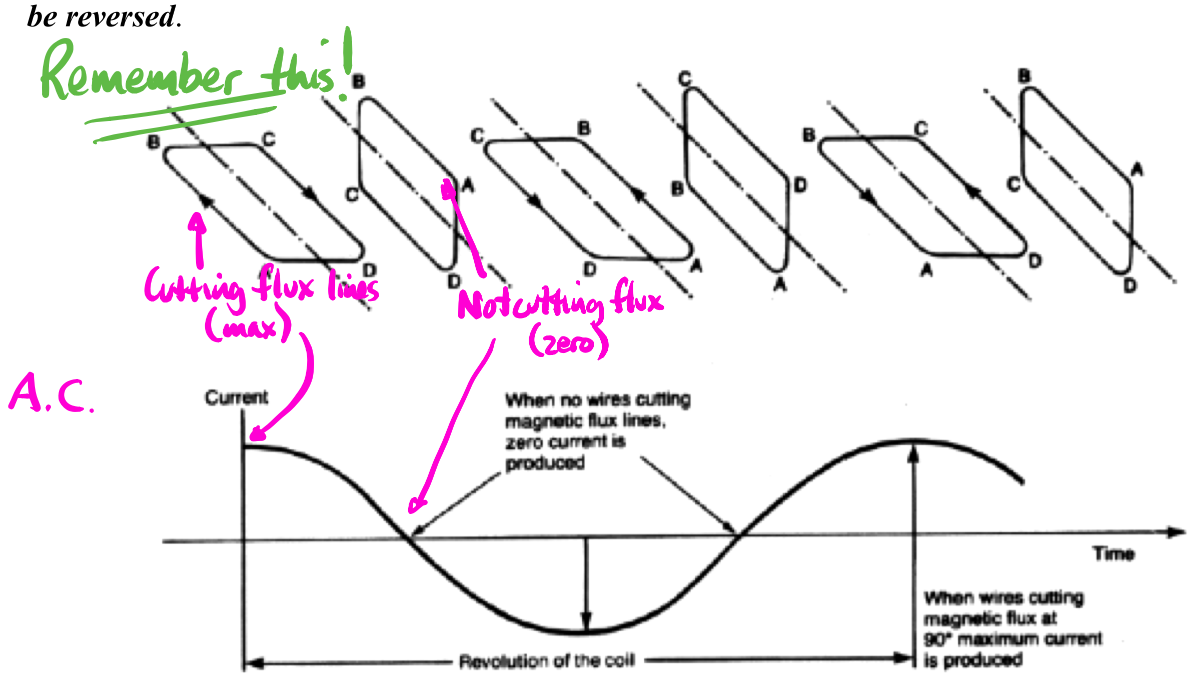


In generators, the coil is turned mechanically (by steam, wind, water or a small petrol/diesel motor).

This induces an EMF that will produce a current flow if the coil is part of a complete circuit.

In the A.C. generator (as seen before), the ***maximum EMF*** occurs when the sides of the coil are cutting the magnetic field at right angles (***when the plane of the coil is parallel to the field***) and it drops to zero 1/4 of a turn later when the sides of the coil are moving parallel to the magnetic field.

Every half a turn, the EMF will return to its maximum value but the ***induced current direction will be reversed***.



The changes in current as a coil is rotated in a magnetic field

Ignore this!

This graph is about the flux threading the loop!

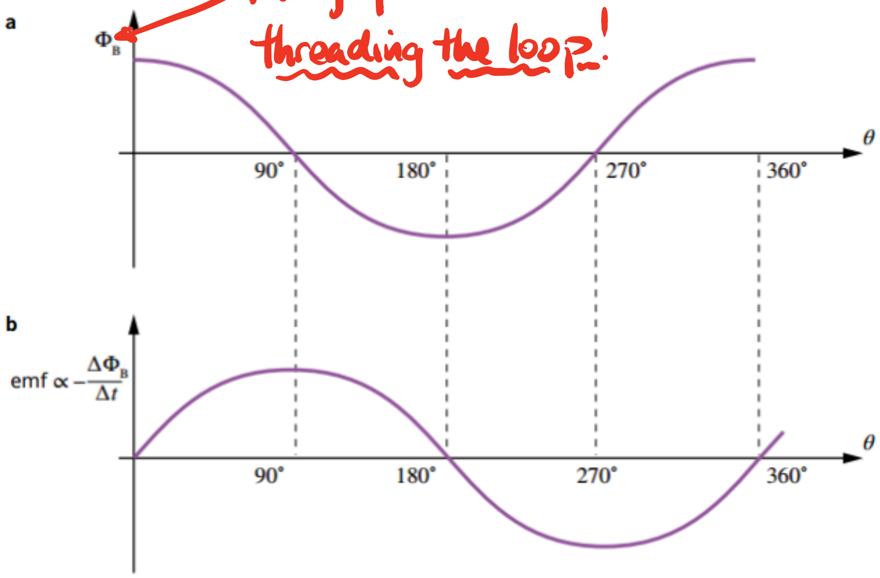


FIGURE 6.4.3 (a) The flux, ϕ, through the loop of Figure 6.4.2 as a function of the angle between the field and the normal to the plane of the area, θ. (b) The rate of change of flux and hence emf through the loop as a function of the angle between the field and the normal to the plane of the loop, θ. The loop is rotating at a constant speed.

Due to the current reversing direction every half a turn, two slip rings or commutators are used to remove the induced current from the coil.

In power stations, the generators operate at 3000 r.p.m. which equates to a frequency of 50 Hz. Hence our power supply has its current direction changing 50 times each second.

Direct Current (D.C.)

Direct current generators work on the same principle except that a ***split-ring commutator*** is used to maintain the direction of flow of current from the coil in one direction.

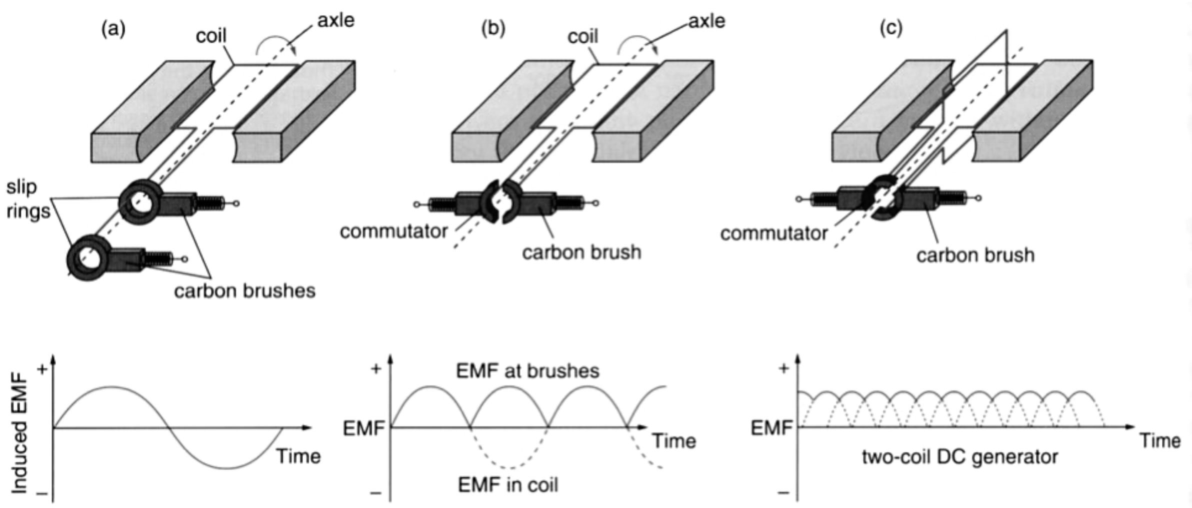


Figure 4.17
(a) An AC generator uses simple slip rings to take the current from the coil. (b) A DC generator has a commutator to reverse the direction of the alternating current every half cycle and so produce a DC output. (c) Two (or more) coils may be used to smooth the DC output.

Alternating Voltage and Current

An AC generator produces a current that varies sinusoidally over time with the change in magnetic flux. In Australia, mains electricity oscillates at 50 Hz and reaches a peak voltage of $\pm$ 340 V.

$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$

(on data sheet)

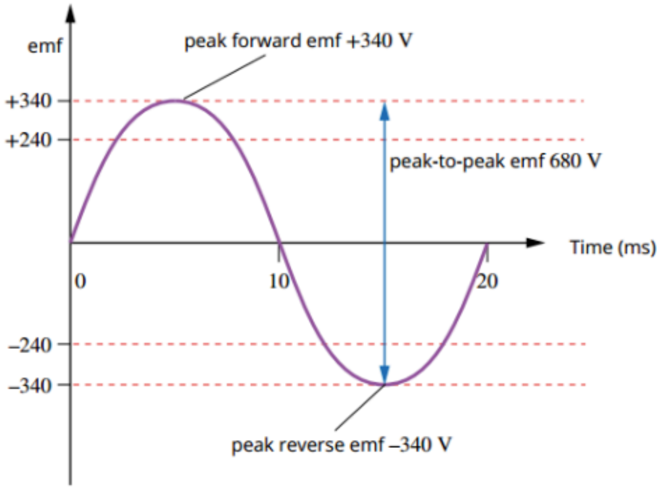


FIGURE 6.4.8 The voltage in Australian power points oscillates between +340 V and –340 V, 50 times each second. The value of a DC supply that would supply the same average power is 240 V.

It is often more useful to know the average power produced in a circuit so that the equations for DC electricity can be used with AC electricity. The average power is obtained by using a value for the voltage and current equal to the peak values divided by √2. This is referred to as the ***root mean square (rms) value***.

EXTENSION

Deriving the root mean square formulae

In an AC circuit, the power produced in a resistor is equal to $\frac{V^2}{R}\sin^2\theta$.

The average power will be given by:

$$\frac{1}{2}\frac{V_{peak}^2}{R}$$

If this same power was to be supplied by a steady (DC) source, the voltage V_{av} of this source would have to be such that:

$$\frac{V_{av}^2}{R} = \frac{1}{2}\frac{V_{peak}^2}{R}$$

Simplifying:

$$V_{av}^2 = \frac{V_{peak}^2}{2}$$

$$V_{av} = \frac{V_{peak}}{\sqrt{2}}$$

This average voltage is known as the root mean square voltage or V_{rms}. It is the value of a steady voltage that would produce the same power as an alternating voltage with a peak value equal to $\sqrt{2}$ times as much.

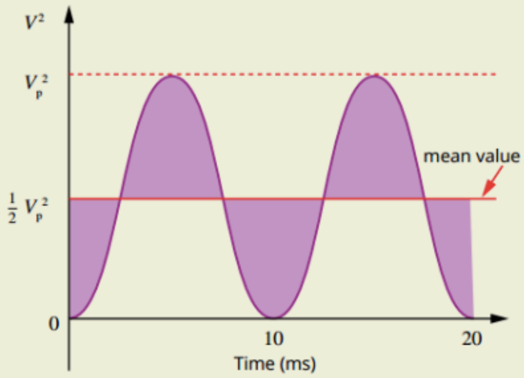


FIGURE 6.4.9 The power transmitted is proportional to the area under a V^2 graph. Clearly, the power transmitted by an AC circuit (with V_{peak}) is the same as that in a DC circuit with a voltage equal to the square root of $\frac{1}{2}V_{peak}^2$, that is $\frac{V_{peak}}{\sqrt{2}}$.

Example

PEAK AND RMS AC VALUES

A 60.0 W light globe is connected to a 240 V AC circuit. Calculate the peak power use of the light globe.

Thinking	Working
Note that the values given in the question represent rms values. Power is $P = VI$ so both V and I must be known to calculate the power use. The voltage V is given, and the current I can be calculated from the rms power supplied.	$P_{rms} = V_{rms} I_{rms}$ $I_{rms} = \frac{P_{rms}}{V_{rms}}$ $= \frac{60.0}{240} = 0.250$ A
Substitute in known quantities and solve for peak power.	$P_{peak} = \sqrt{2}V_{rms} \times \sqrt{2}I_{rms} = 2V_{rms}I_{rms}$ $= 2 \times V_{rms} \times I_{rms}$ $= 2 \times 240 \times 0.250$ $= 120$ W

$V_{peak} = \sqrt{2}\, V_{rms}$

PHYSICS IN ACTION

Back emf in DC motors

The description of the construction and operation of a generator shows that a DC motor and a generator share a lot in common and can function in either way; so every motor can also be used as a generator. The motors of electric trains, for instance, work as generators when a train is slowing down, converting kinetic energy to electrical energy and putting it back into the electrical supply grid. Regenerative brakes in cars work in a similar way. A DC motor will also generate an emf when running normally. This is termed the '**back emf**'.

The back emf generated in a DC motor is the result of current produced in response to the motion of a conducting coil inside the motor in the presence of an external magnetic field. This back emf, according to Lenz's law, opposes the change in magnetic flux that created it, so this induced emf will be in the opposite direction to the applied emf creating the motion in the first place. The net emf used by the motor is thus always less than the applied voltage:

$$\varepsilon_{net} = V - \varepsilon_{back}$$

As the motor increases speed, the induced back emf will also increase, thus reducing the net emf and therefore reducing the applied current in the motor coils. This has the effect of reducing the force on the motor and so it will accelerate the motor less and less. Eventually the back emf will equal the applied emf and, theoretically, there will be no net emf and no current flowing in the coil. As the current causes the force on the wire there will be no force and consequently no acceleration; the motor is at its maximum speed.

When a load is applied to the motor, such as a when a drill is drilling into a piece of wood, the speed of the motor will generally reduce. This will reduce the back emf and increase the net emf and hence the current in the motor. This increase in current causes a force on the motor coil that acts to do work on the wood. If the load brings the motor to a sudden halt—say, an electric drill bit getting stuck in some wood—the sudden decrease in back emf and increase in net emf will cause the current to increase rapidly such that it may be high enough to burn out the motor and the motor windings. To protect the motor, a resistor is placed in series. It is switched out of the circuit when the current drops below a predetermined level and is switched back into the circuit for protection once the level is exceeded.

Regenerative braking

Every time a traditional car brakes, it is wasting energy. When the car slows down, the kinetic energy that was propelling it forwards is largely lost as heat. That energy, and the original fuel that was used to create it, is essentially wasted. **Regenerative braking** overcomes this problem by recapturing much of the car's kinetic energy and converting it to electrical energy to recharge the car's battery, as shown in Figure 6.4.7.

Regenerative braking is nothing new. Trams have been utilising regenerative brakes for many years, feeding excess energy back into the electrical supply grid. With more focus on energy efficiency and hybrid cars, regenerative brakes are now becoming more common in cars.

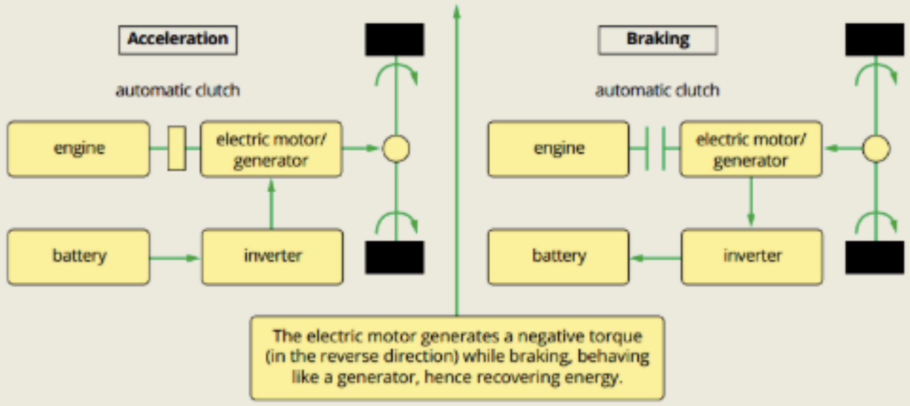
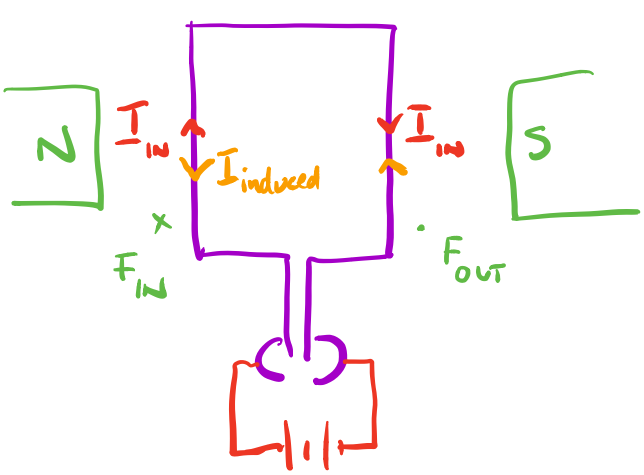


FIGURE 6.4.7 A regenerative braking system, showing the flow of electrical energy during acceleration and braking.

DC MOTOR



$I_{induced}$ opposes I_{input}

$\therefore$ I in the coil decreases.

PHYSICS IN ACTION *(continued)*

Traditional brake systems rely on a brake pad producing friction against a metal rotor or disc to slow or stop the vehicle, with additional friction to bring the vehicle to a full stop coming from the action between tyres and road. Friction turns the car's kinetic energy into heat. With induction-based regenerative brakes, it's actually the same system that propels the car that does most of the braking, with the kinetic energy being recovered to be used again.

Regenerative braking is largely used in electric and hybrid cars that make use of electric motors to propel the car and have batteries to store charge. The braking system runs the car's electric motor in reverse, converting mechanical energy into electrical energy and then into the car's batteries for storage.

During acceleration, the system uses electricity from the battery, converts it to the voltage required via an inverter transformer and sends it to the electric motor to power the car. During braking, the process is reversed. Mechanical energy from the moving car is converted by the motor into electrical energy and fed back through the inverter for storage in the car's batteries.

Specialised electronic circuits determine when the motor should reverse and direct the electricity produced back to the batteries or, in some cases, to a series of capacitors for shorter term storage. The process also provides very effective braking via the eddy currents induced to oppose the change in flux.

PHYSICS IN ACTION

Three-phase generators

Many industrial applications require a more constant maximum voltage than is possible from a single coil. These applications require a three-phase power supply. The coils are arranged such that the emfs vary at the same frequency, but with the peaks and troughs of their waveforms offset to provide three complementary currents with a phase separation of one-third of a cycle, or 120°. The resulting output of all three phases maintains an emf near the maximum voltage more continuously. Standard electrical supplies include three phases, but most home applications only require a single phase to be connected.

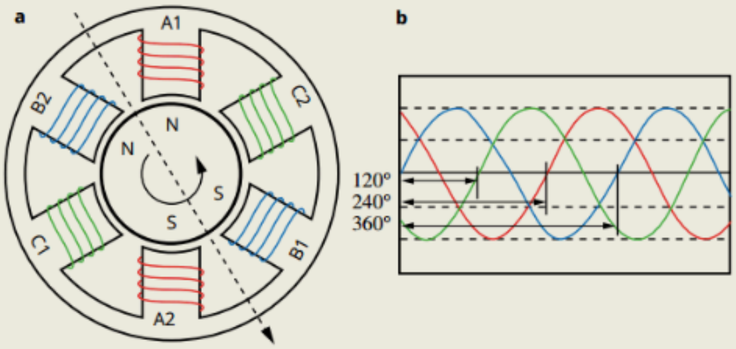


FIGURE 6.4.5 (a) A three-phase power supply has three coils, each producing an output 120° out of phase with the adjoining coil. (b) The resulting output can be combined for a more constant supply voltage.

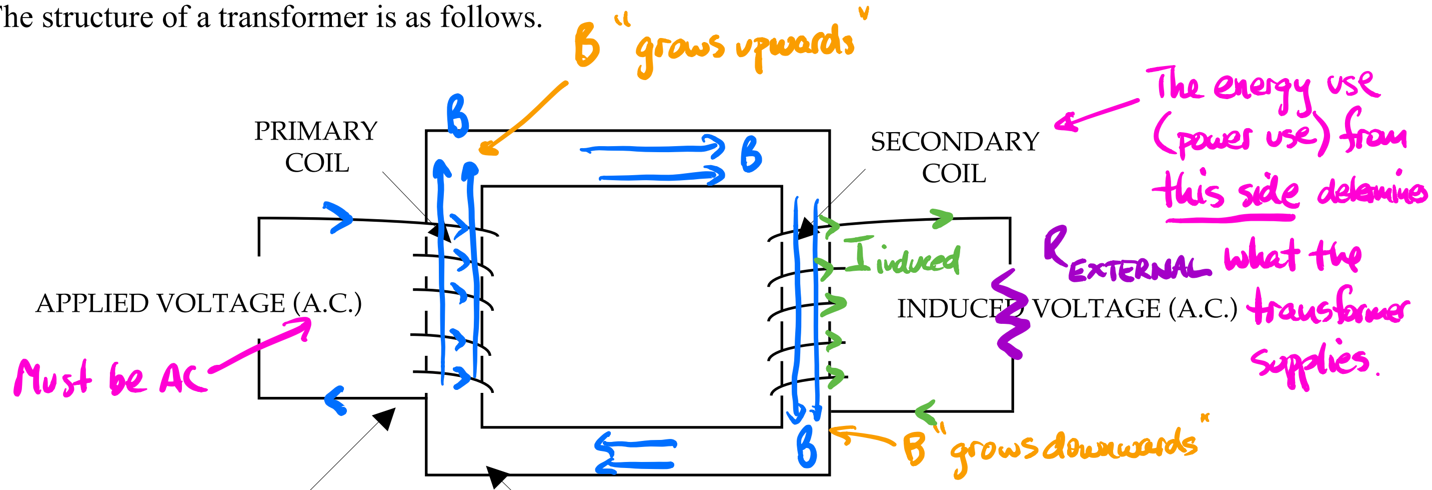
TRANSFORMERS

From Faraday's formula:

$$\boxed{EMF = -N\frac{\Delta \Phi}{\Delta t}}$$

it is possible to produce an induced EMF not only by having a conductor or coil moving in a constant magnetic field, but also to ***have the conductor stationary in a changing magnetic field***. This is the principle behind the operation of a transformer.

The structure of a transformer is as follows.



COMMON SOFT IRON CORE CONCENTRATES THE MAGNETIC FLUX AND INCREASES THE EFFICIENCY OF THE TRANSFORMER.

VARYING THE CURRENT IN THE PRIMARY COIL CHANGES THE MAGNETIC FLUX THROUGH THE SECONDARY COIL. THIS INDUCES AN EMF IN THE SECONDARY COIL.

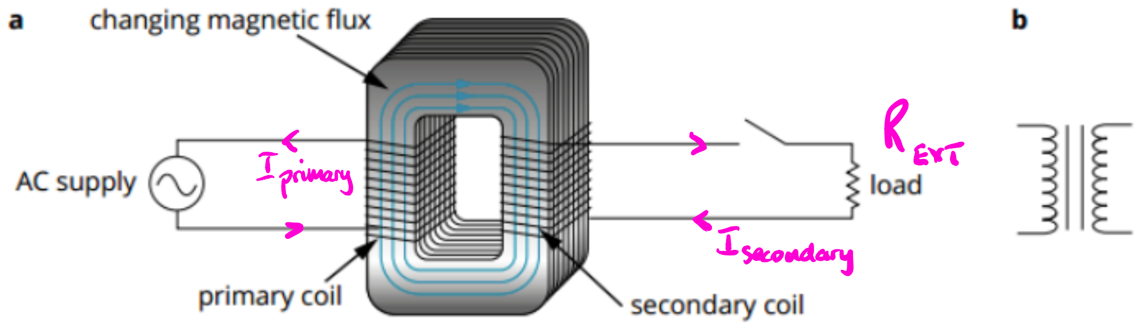


FIGURE 6.4.10 (a) In an ideal transformer, the iron core ensures that all the flux generated in the primary coil also passes through the secondary coil. (b) The symbol used in circuit diagrams for an iron-core transformer.

The easiest way to alter the current in the primary coil is to use AC current that constantly changes direction.

This induces an AC current in the secondary coil.

$EMF = -\frac{N\Delta\phi}{\Delta t} = -\frac{N\Delta BA}{\Delta t}$

i.e. The secondary coil is ***stationary*** in a ***changing magnetic field***.

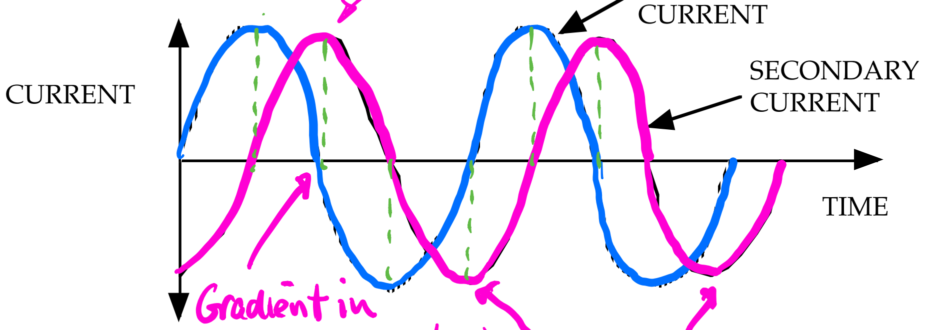
From the current sine wave produced by an A.C. generator, ***the current is changing at its maximum rate as it passes through zero while changing direction***.

In the transformer, this corresponds to the ***maximum change in the magnetic field*** associated with the current in the primary coil.

Consequently, the ***EMF induced in the secondary coil is at a maximum at this time***.

i.e. There is a 90° ($\frac{\pi}{2}$) phase difference between the primary current and secondary current.

$\frac{\pi}{2}$ phase lag in the secondary coil.



Gradient in primary is greatest
⇒ Induced current in secondary is max.

The relationship between the number of turns and voltages in the primary and secondary coils is given by:

$$\boxed{\frac{V_S}{V_P} = \frac{N_S}{N_P}}$$

On data sheet

where V_p = voltage in the primary coil (V)
V_s = voltage in the secondary coil (V).
N_p = number of turns per unit length in the primary coil.
N_s = number of turns per unit length in the secondary coil.

Note
The use of DC current in the primary coil will ***not*** induce an EMF in the secondary coil since the magnetic field will not change.

But it will change if the primary current is turned on and off.

From the above equation, the following is true.

Step-up transformer: $N_s > N_p$ so $V_s > V_p$

Step-down transformer: $N_s < N_p$ so $V_s < V_p$

For a transformer, the ***power dissipated in the primary coil is essentially equal to the power delivered to the secondary coil***.

i.e. ***For calculations, transformers are assumed to be 100 % efficient***.

If it isn't, you need to factor it in.

From this can be calculated:

$$P\ (\text{primary}) = P\ (\text{secondary})$$

$$\Rightarrow \quad V_p I_p = V_s I_s$$

$$\boxed{\frac{V_s}{V_p} = \frac{I_p}{I_s}}$$

Hence the relationships within a transformer can be summarised as follows.

$$\boxed{\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}}$$

Learn this!

Thus, when ***voltage is stepped up, the current is stepped down***, and there is no power gain as a result of transformer action.

As with all machines there is some energy loss.

(See notes on Power Output.)

Step-down

Example 1

N_p I_p

A transformer has a primary winding of 200 turns that carries a current of 5.00 A at an input voltage of 48.0 V. The secondary winding has 50 turns.

V_p N_s

(a) Calculate the voltage output from the transformer.

(b) Assuming the transformer to be 100% efficient, what is the power output of the transformer?

(a)

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

$$\Rightarrow \frac{V_s}{(48.0)} = \frac{(50)}{(200)}$$

$$\Rightarrow V_s = 12.0 \text{ V}$$

The voltage output = 12.0 V .

(b) Given that the transformer is 100% efficient:

$$\begin{aligned} P_{secondary} &= P_{primary} \\ &= V_p I_p \\ &= (48.0)(5.00) \\ &= 2.40 \times 10^2 \text{ W} \end{aligned}$$

The power output = 2.40×10^{2} W.

Example 2

V_s V_p N_s

step-down by ×20

A transformer used for charging a mobile phone battery operates on 12.0 V from the mains 240.0 V supply.

a If the number of turns in the secondary coil is 100, what will be the number of turns in the primary coil?

b If the battery charging requires a current of up to 4.00 A, what will be the current and the power drawn from the mains?

I_s

Solution

a

$N_s = 100$

$V_s = 12.0$ V

$V_p = 240.0$ V

$$\frac{N_p}{N_s} = \frac{V_p}{V_s}$$

$$N_p = \frac{N_s V_p}{V_s} = \frac{(100)(240)}{12.0}$$

$$= 2000$$

b The current ratio is the inverse of the turns and voltage ratio so that the current in the primary will be:

$N_s = 100$

$I_s = 4.00$ A

$V_p = 240.0$ V

$V_s = 12.0$ V

$I_s = 4.00$ A

$$\frac{N_p}{N_s} = \frac{I_s}{I_p}$$

$$I_p = \frac{I_s N_s}{N_p} = \frac{(4.00)(100)}{2000}$$

$$= 0.200 \text{ A}$$

$$P = VI$$

$$= (12.0)(4.00)$$

$$= 48.0 \text{ W}$$

The power required by the battery was 48.0 W. Alternatively, the input power was 48.0 W, which is of course the same, as we assumed that the transformer itself uses no power.

Power Output

Good quality transformers are usually 98-99 % efficient with 1-2 % of the energy being wasted as heat.

For this reason, electrical appliances that involve a transformer (e.g. TV, computer screens, CRO) are ventilated at the rear so this “lost heat” can escape from the machine and not cause any damage due to heat build-up.

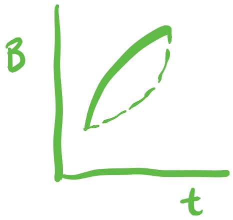
The three main reasons for these heat losses are as follows
.

Remember $R = \frac{\rho l}{A}$

(a) ***Copper losses*** - the resistance of the copper wires in the primary and secondary coils (resistivity) cause heat losses according to the formula $P = I^2R$.

whichever coil has the greatest I must have the thicker wire to decrease R.

(b) ***Eddy current losses*** - as explained below.



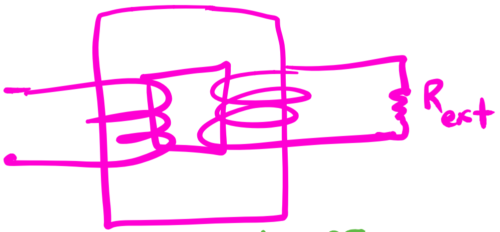
(c) **Hysteresis** - the magnetisation of the core is repeatedly reversed by the alternating magnetic field. The repeating core magnetisation process expends energy and this energy appears as heat. The heat generated can be kept to a minimum by using a magnetic material that has a low hysteresis loss. Hence, ***soft iron is often chosen*** for the core material because the magnetic domains within it changes rapidly with low energy loss.

Example

A step-down transformer has 100 turns in its primary coil (N_p) and 25 turns in its secondary coil (N_s). The power of the transformer is 3.00 W ($P_p = P_s$) and the voltage output is 6.00 V (V_s).

Assume 100% efficiency.

(a) Calculate the current in the primary coil.
(b) Explain which coil will have the thickest wire and why.



$N_p = 100$ $\quad N_s = 25$
$V_p = ?$ $\quad V_s = 6.00V$
$P_p = 3.00W$ $\quad P_s = 3.00W$

(a)

$$\frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$\Rightarrow \frac{25}{100} = \frac{6.00}{V_p}$$

$$\Rightarrow V_p = 24.0 \text{ V}$$

$$P_{primary} = V_p I_p$$

$$\Rightarrow I_p = \frac{P_p}{V_p} = \frac{3.00}{24.0} = 0.125 \text{ A}$$

$$\therefore \underline{I_p = 0.125 \text{ A}}$$

(b) The power losses in the coils is given by $P = I^2R$.

Comes from $R = \frac{\rho l}{A}$ $\Rightarrow R \propto l$, $R \propto \frac{1}{A}$

From this relationship, the largest current will have the biggest power loss.

To minimise this, the resistance of that wire can be made smaller by increasing its thickness.

Hence the **coil with the greatest current must have the thicker wire** .

By inspection, the current in the secondary coil is the greatest.

∴ **The secondary coil is made of the thicker wire** .

Eddy Currents

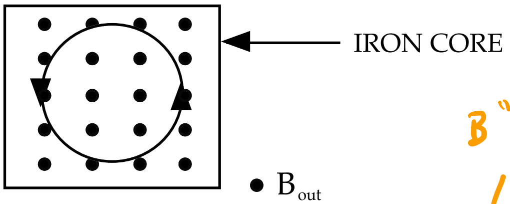
These are circulating currents of electricity that can be set up in conductors (particularly if they are large and plate-like) either moving in a constant magnetic field or stationary in a changing magnetic field.

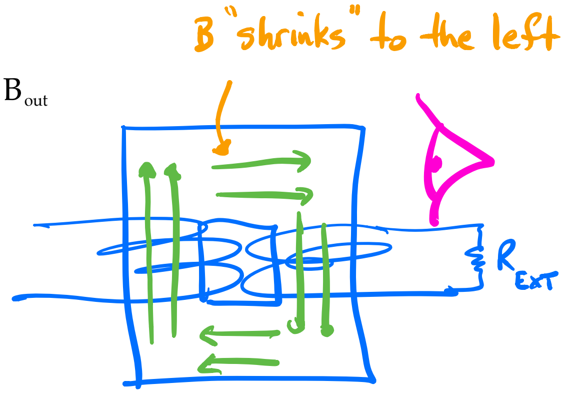
These induced currents are closed loops that circulate in planes perpendicular to the magnetic flux.

The magnetic field will "shrink" into the page as the magnetic field changes direction.

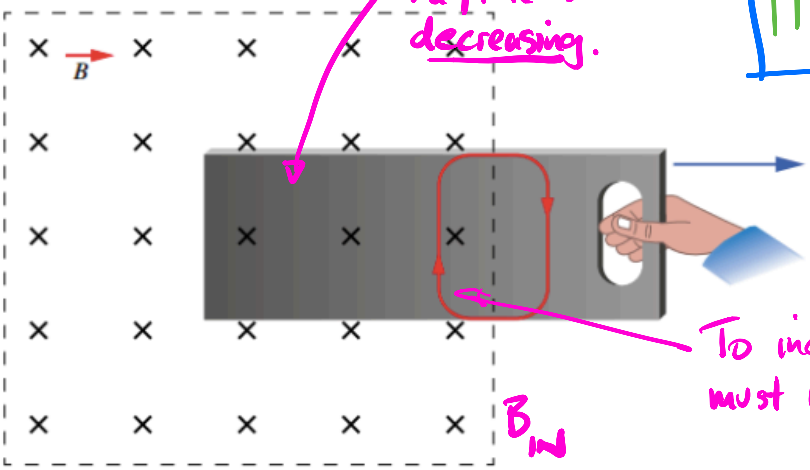
Using the Right Hand Grasp Rule. the thumb is out of the page to oppose this "shrinkage".

The fingers curl around in an anticlockwise direction.





Φ through the plate is decreasing.



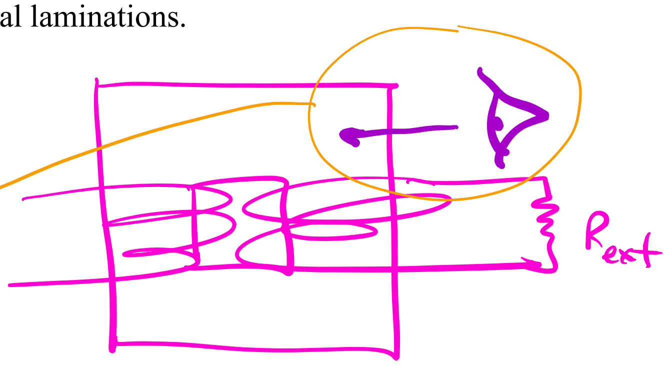
To increase Φ, thumb must be into the page

FIGURE 6.3.6 As the metal plate is moved towards the right, out of the magnetic field that is directed into the page, an eddy current forms in a clockwise direction. This eddy current resists the motion of the plate.

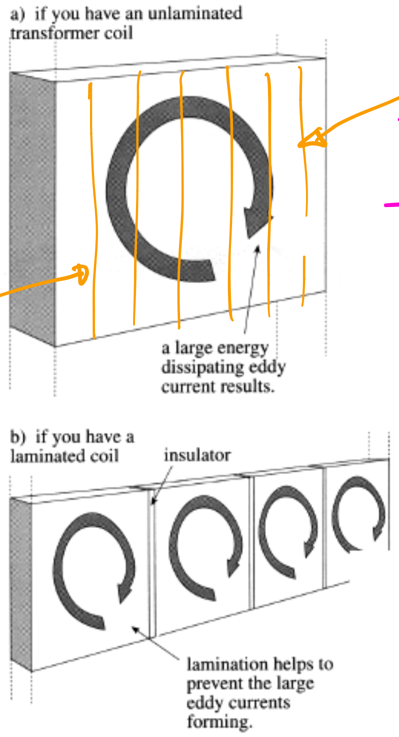
Eddy currents will produce a magnetic field that will cause a force to oppose the motion inducing the EMF (in motors and generators) and/or dissipate energy as heat (in transformers or electromagnetic braking systems).

Transformers have ***laminated iron cores*** (made up of thin sheets) to minimise the eddy currents produced. Thus, the amount of heat generated is small.

The planes of these laminations are set in planes ***parallel to the magnetic flux*** so that the eddy currents are confined to the width of the individual laminations.



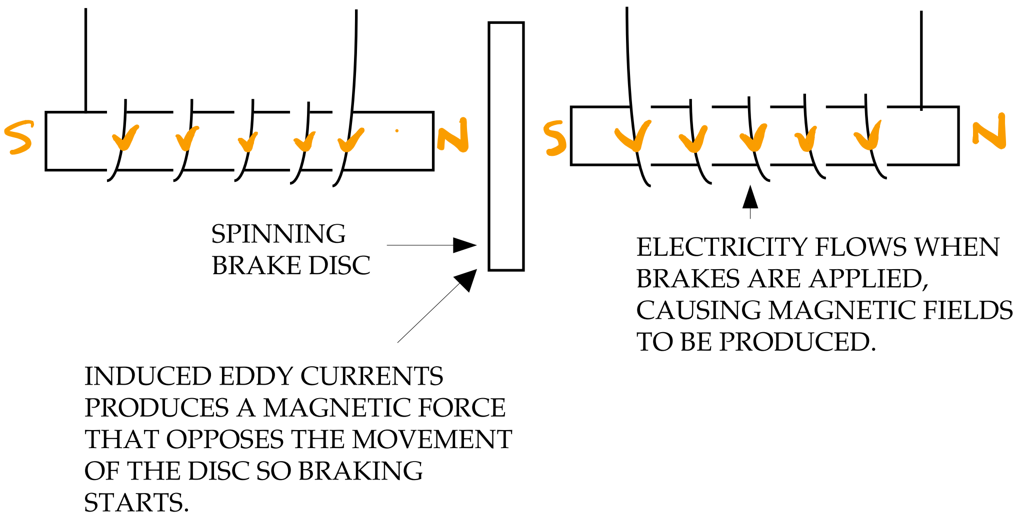
laminate vertically to minimise eddy currents.



By having smaller currents, the heat losses are smaller.

Collectively the total energy is smaller than that produced by one large eddy current.

Magnetic braking systems rely on the production of large eddy currents so that their magnetic fields can interact with the permanent magnetic field present to oppose the motion of the metal brake disc and slow the vehicle.



As the eddy current is large, large amounts of heat are generated in these braking systems.

POWER TRANSMISSION

Domestic electricity is Australia is usually generated a great distance from the areas of greatest demand within large population areas.

e.g. Muja power station near Collie supplies the majority of electricity to the Perth metropolitan area some 200 km away.

Due to the great distances involved, there is a large amount of resistance within the transmission lines. The movement of current through this resistance will cause a heating effect and hence a loss of energy.

The aim is to minimise this heat loss from the grid system by having as ***low a current as possible flowing***.

$\uparrow P_{loss} = I^2R$

From Year 11 work, the power produced by a current is given by:

$$\boxed{P = VI = I^2R = \frac{V^2}{R}}$$

$\therefore I$ low $\Rightarrow P_{loss}$ low

The power produced in a generator at a power station can be considered to be ***constant***.

From P = VI, in order to have a small current, ***V must be high***.

Hence the output voltage from a power station is usually stepped-up to 330 kV or 500 kV, then stepped-down as it reaches the areas where it is needed.

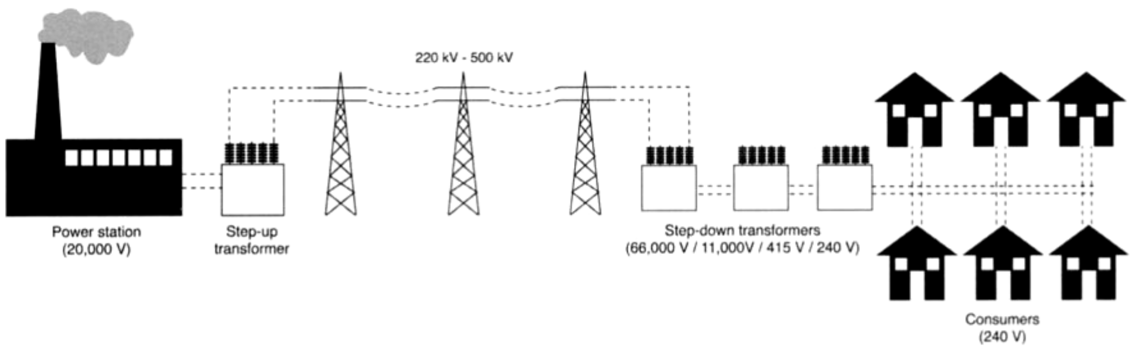
Note
From P = I^2R, it can be seen that the power loss in the system is proportional to I^2.

The smaller I is, the smaller the power loss due to heat.

From P = VI:

every 10-fold increase in V means a 10-fold decrease in I (since the power P remains constant).

=> a 100-fold decrease in the power loss P (since P = I^2R).



Just be aware of this!

ENVIRONMENTAL IMPACT OF ELECTRICITY PRODUCTION AND TRANSMISSION

In Australia, most electricity is generated by burning coal (other sources are burning gas, hydroelectric schemes, wind power and solar cells).

The burning of coal is not very efficient as a lot of heat (≈ 50 %) is wasted. This leads to thermal pollution of the air and water streams that are used to produce the steam to drive the turbine generators.

Gases from the burning of coal (CO_2, SO_2, oxides of nitrogen) add to the air pollution problems in the atmosphere. Not only do they contribute to the Greenhouse Effect, they can all form acid rain by dissolving into rainwater. This acid solution has killed large areas of forest and crops in North America and Europe, chemically erodes buildings and statues by reacting with the stone, and killed aquatic life in lakes and streams. SO_2 and the oxides of nitrogen are also irritants to the respiratory system and cause problems for the elderly and asthmatics.

The pylons used to hold transmission lines need to be large and strong by necessity. However, they are a form of visual pollution for many people.

Due to the high current that flows (≈ 600 A), a strong electromagnetic field is produced. There has been some concern in recent times that people living near these lines are more prone to certain types of cancers. The research into this area is continuing.

The very high voltages used in transmission cause an atmospheric effect called ***corona phenomena***. The air breaks down around the conductors and conducts with a voltage gradient of 800 Vmm^{-1} of air. Hence the lines are kept at least 700 mm from the pylons by insulators to avoid arcing.

This air leakage creates additional power losses and affects TV, cellular phone and radio communications, causes visual coronas and audible crackling. Such effects are most prevalent when the air is moist.

CALCULATIONS INVOLVING POWER

The equations that are used are:

$$\boxed{P = VI = I^2R = \frac{V^2}{R}}$$

When considering calculations involving power from a power station, remember that:

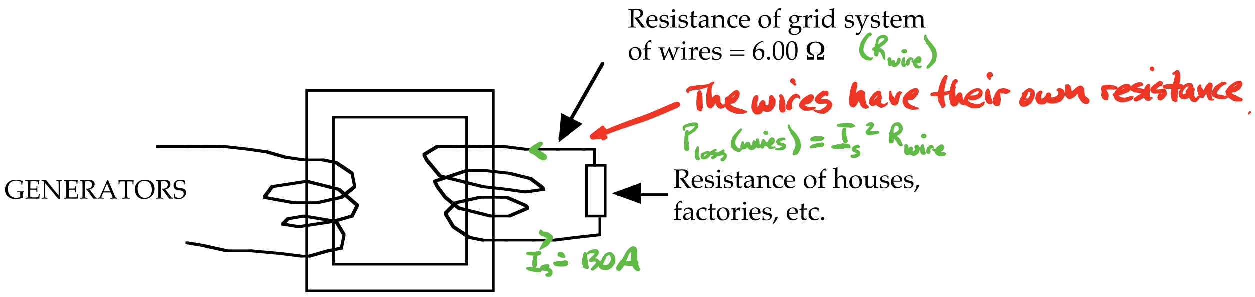
1. the power of the individual generators adds together to give the overall power output.
   e.g. four generators each of 15.0 MW will produce 60.0 MW output.

2. the power output from the station is determined by the power used by each of the houses, factories, buildings, etc. on the grid system.

Work from house back to power station.

Example 1
A power station generator, rated at 65.0 MW, produces a voltage of 11.0 kV that is stepped-up to 5.00×10^2 kV. The grid system for that area has a combined resistance of 6.00 Ω. Calculate:

(a) the current induced in the secondary coil of the transformer.
(b) the current produced in the grid system.
(c) the power loss that occurs in the grid system.



$V_p = 11.0$ kV $V_s = 5.00 \times 10^2$ kV

$P_p = 65.0$ MW $P_s = 65.0$ MW ← Assumes 100 % efficiency.

$I_p = ?$ $I_s = ?$

(a) For the transformer:

$$P_{primary} = P_{secondary}$$
$$\text{i.e}\quad P = V_pI_p = V_sI_s$$
$$\Rightarrow I_s = \frac{P_s}{V_s}$$
$$= \frac{6.50 \times 10^7}{5.00 \times 10^5}$$

Convert

$$= 1.30 \times 10^2 \text{ A}$$

∴ The induced current = 1.30×10^2 A.

(b) For the grid system, the current induced in the secondary coil must equal the current seen in the external resistance of the grid system.

Hence the current in the grid system = 1.30×10^{2} A.

(c) To calculate the power loss:

$$P = I^2R$$
$$= (1.30 \times 10^2)^2(6.00)$$
$$= 1.014 \times 10^5 \text{ W}$$

$\therefore$ The power loss = 1.01×10^{5} W.

$P_{loss}(\text{wires}) = I_s^2 R_{wire}$
$= (130)^2(6.00)$
$= 1.00 \times 10^5$ W
$= 0.100$ MW

Note

The above calculations are very simplistic and really only apply to ***D.C. current***.

A.C. current is much more easily generated and transformed, so it is the preferred current. This means that power losses are really due to impedance and inductance in the grid system. Calculations based on these concepts are difficult to understand and so have been avoided!

Example 2

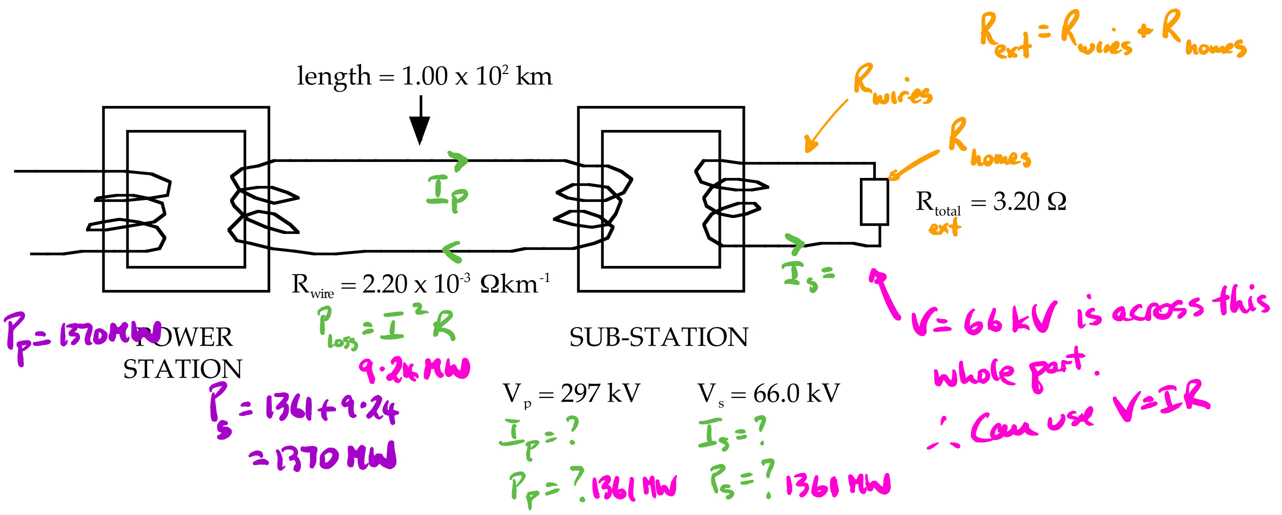
A power station sends electricity through power lines 1.00 x 10^2 km long and resistance 2.20 x 10^{-3} Ωkm^{-1} to a sub-station. Here 297 kV is stepped-down to 66.0 kV for transmission to a large part of the metropolitan area. The total resistance of this area (transmission wires, homes, factories, etc.) is 3.20 Ω.

(a) Calculate the power used by the area serviced by the sub-station.

(b) Determine the power loss occurring along the transmission lines between the sub-station and the power station.

(c) Calculate the power required to be generated by the power station.

(d) What would happen if the total power required by the metropolitan area was greater than the maximum that could be generated by the power station?



(a) To calculate the current flowing from the sub-station:

$$V_s = I_s R_{total}$$

$$\Rightarrow \quad I_s = \frac{V_s}{R_T} = \frac{66.0 \times 10^3}{3.20} = 2.062 \times 10^4 \text{ A}$$

The power used on the secondary side of the sub-station transformer is given by:

$$P_s = V_s I_s = (66.0 \times 10^3)(2.062 \times 10^4) = 1.361 \times 10^9 \text{ W}$$

$\therefore$ Power used by the area = 1.36 x 10^9 W

$$P_s = \frac{V_s^2}{R_{ext}} = \frac{(66.0\times10^3)^2}{3.20} = 1.36\times10^9 \text{ W} = 136 \text{ MW}$$

Should be 100% efficient in WACE.

(b) The power produced in the primary coil of the sub-station = 1.36 x 10^{9} W (assuming 100 % efficiency in the transformer).

To calculate the current flowing in the primary coil (and hence the transmission lines):

$$P_p = V_p I_p$$
$$\Rightarrow \quad I_p = \frac{P_p}{V_p}$$
$$= \frac{1.361 \times 10^9}{297 \times 10^3}$$
$$= 4.582 \times 10^3 \text{ A}$$

There are **two wires** that go between the power station and the sub-station.

$$\therefore \quad R_{wire} = 2(1.00 \times 10^2)(2.20 \times 10^{-3}) = 0.440\ \Omega$$

$$P_{loss} = I^2R$$
$$= (4.582 \times 10^3)^2(0.440)$$
$$= 9.238 \times 10^6 \text{ W}$$

$\therefore$ Power loss = 9.24 x 10^6 W   9.24 MW

(c) Power drawn on the secondary side of the power station transformer is given by:

$$P_{station} = P_{sub\text{-}station} + P_{loss}$$
$$= 1.361 \times 10^9 + 9.238 \times 10^6$$
$$= 1.370 \times 10^9 \text{ W}$$

$\therefore$ Power required = 1.37 x 10^9 W

(d) The power that is drawn from the power station transformer is determined by the total power used up by the consumers, as well as any power losses along transmission lines.

There is a maximum value for the amount of power that can be generated by a power station.

If the demand is greater than this maximum, then the station will shut down its generators to avoid damaging them.

To avoid this, the amount of power being demanded is closely monitored. If it approaches the maximum value, "power rationing" will occur and various suburbs will be "blacked out" to avoid over-loading the system.

Example 3

An ac generator produces 10 kW of power which is to be transmitted at 240 V down a line 5 km long and whose total resistance is 1.5 Ω. Calculate

(a) **the power lost in the line;**

(b) **the voltage which reaches the end of the line.**

$P_{loss} = I_s^2 R = 2.6\text{kW}$

$R_{wire} = 1.5\,\Omega$

$V_s = 240\text{V}$

$P_s = 10\text{kW}$

$I_s = ?$ 41.7A

$P_{end} = 10 - 2.6 = 7.4\text{kW}$

$I_s = 41.7\text{A}$

$V_{end} = ?$

(a)

$P = 10^4$ W

$V = 240$ V

$I = ?$

$R = 1.5\,\Omega$

Since $P = VI$

$$I = \frac{P}{V} = \frac{10000}{240} = 41.7 \text{ A}$$

$\therefore$ Power lost $P = I^2 R$ $\quad P_{loss} = I_s^2 R$

$$= (41.7)^2\,(105)$$

$$= 2.60 \times 10^3 \text{ W}$$

$$= 2.6 \text{ kW}$$

This represents over ¼ of the power being lost down the line.

(b)

$V = ?$

$I = 41.7$ A

$R = 1.5\,\Omega$

Voltage lost along the line

$$\left.\begin{aligned} V &= IR \\ &= (41.7)\,(1.5) \\ &= 62.6 \text{ V} \end{aligned}\right\} \quad V_{drop} = I_s R_{wire}$$

Voltage which reaches the end of the line will be

$$V_{end} = \frac{P_{end}}{I_s}$$

$$V(\text{end}) = 240 - 62.6 = \underline{177 \text{ Volts}}$$

or

$$\therefore P_{end} = V_{end} \times I_s$$

$$7.4 \times 10^3 = V_{end} \times 41.7$$

$$V_{end} = 177 \text{ volts}$$

This is not a very satisfactory state of affairs. Imagine the loss over a much longer line. The solution is high voltage transmission.

Example 4

At a time of peak demand 835 MW is to be provided to Perth from the Muja Power Station, along a power line with a total resistance of 2.00 Ω.

a What would be the total transmission power loss if the potential at Perth was to be:
  i 240 V? **ii** 500 kV?

b What potential would be needed at the Muja Power Station end of the line to achieve these potentials at Perth?

c How would the answers to parts a (ii) and b change if the resistance of the power line was halved?

Solution

a i

$P_{Perth} = 635 \times 10^6$ W $\qquad I = \frac{P_{Perth}}{V_{Perth}} = \frac{635 \times 10^6}{240}$

$V_{Perth} = 240$ V $\qquad = 2.65 \times 10^6$ A

Thus the power loss would be:

$R = 2.00\ \Omega \qquad P_{loss} = I^2R = (2.65 \times 10^6)^2(2.00)$

$I = 2.65 \times 10^6$ A $\qquad = 1.40 \times 10^{13}$ W

This is actually much greater than the power supplied, so 99.995% of the power is lost in transmission!

ii

$P_{Perth} = 635 \times 10^6$ W $\qquad I = \frac{P_{Perth}}{V_{Perth}} = \frac{635 \times 10^6}{500 \times 10^3}$

$V_{Perth} = 500 \times 10^3$ V $\qquad = 1.27 \times 10^3$ A

Thus the power loss would be:

$R = 2.00\ \Omega \qquad P_{loss} = I^2R = (1.27 \times 10^3)^2(2.00)$

$I = 1.27 \times 10^3$ A $\qquad = 3.23 \times 10^6$ W

Now 0.505% of the power is lost in transmission!

b To determine the voltage at the supply end we need to calculate the voltage drop along the transmission line:

$I = 2.65 \times 10^6$ A $\qquad \Delta V_{Muja\text{-}Perth} = IR = (2.65 \times 10^6)(2.00)$

$R = 2.00\ \Omega \qquad \Delta V_{Muja\text{-}Perth} = 5.29 \times 10^6$ V

$$V_{Muja} = \Delta V_{Muja\text{-}Perth} + V_{Perth}$$
$$= 5.29 \times 10^6 + 240$$
$$= 5.29 \times 10^6 \text{ V}$$

This is 5.29 million volts—totally out of the question of course!

$I = 1.27 \times 10^3$ A $\qquad \Delta V_{Muja\text{-}Perth} = IR = (1.27 \times 10^3)(2.00)$

$R = 2.00\ \Omega \qquad \Delta V_{Muja\text{-}Perth} = 2.54 \times 10^3$ V

$$V_{Muja} = \Delta V_{Muja\text{-}Perth} + V_{Perth}$$
$$= 2.54 \times 10^3 + 500 \times 10^3$$
$$= 5.03 \times 10^5 \text{ V}$$

This is 503 000 volts—much more achievable.

c As the power loss is directly proportional to the resistance, the power loss (at 500 kV) would halve to 1.61×10^6 W. The voltage drop would also halve and so the supply voltage would be 1270 V less. These are not particularly large differences and as the cost of the power line would increase considerably (twice as much metal, and thus stronger towers needed) it would probably not be worthwhile.

You might like to show for yourself that the diameter of a 1.00 Ω aluminium conductor 200 km long needed to carry the power in this example would be about 8.2 cm. You need to know that the resistivity, ρ, of aluminium is 2.65×10^{-8} Ω m and that $R = \rho L/A$.

Applications

High-voltage DC

Physics in action — High-voltage DC power transmission and the 'base load' question

The limit on the voltage of an electrical power transmission line is the corona effect—the point at which the intense electric field around the cable breaks down the insulating properties of the air. This limit is around 800 kV, which is a little over the peak voltage reached by a 500 kV (RMS) power line. However, the voltage on an AC line only reaches this briefly twice each cycle and so much of the cycle is 'wasted' in terms of carrying current. On the other hand, if a steady DC voltage of 800 kV was used, the effective current, and hence power carried, would be considerably greater—over twice as much in fact.

The big advantage of AC transmission over DC transmission is that AC voltages can easily be changed by transformers. This is particularly important for distributing power around a city where only the last kilometre or so can be at 240 V. In order to change the voltage of DC, it has been necessary to convert it to AC (using an *invertor*), put it through a transformer and then convert it back to DC (with a *rectifier*). However, because of developments in semiconductor technology, which have made high-current, high-voltage thyristors (a type of transistor) available, it is now possible to change high-voltage AC into high-voltage DC and vice versa with very little loss of power. This means that it has become practicable to use high-voltage DC (HVDC) transmission lines.

There are several advantages in using HVDC over AC for long transmission lines. The first was mentioned above—more than twice the power can be sent along the same size cables. Second, any power line has capacitance, which means some current is 'lost' in charging up the line. This happens every cycle for AC, but not at all for DC. There are significant losses associated with this once a power line is more than a few hundred kilometres long. Third, because of the varying current in an AC line, it is surrounded by a varying magnetic field, which we know will induce currents in any conductor nearby. This is why high-voltage AC power lines need to be so far above the ground and why they can't be put underground or under the sea for any distance. It also means power is lost by electromagnetic radiation from the lines. All these factors limit AC transmission lines to a few hundred kilometres.

Figure 4.31
At the heart of an AC-DC converter station are valves comprising power thyristors, which are basically large high voltage transistors.

While the cost of the inverting and rectifying equipment is relatively high, because HVDC transmission does not suffer the problems of AC, it becomes more economical for large distances—up to thousands of kilometres. In fact, losses of only 3% per 1000 km have been achieved.

Another advantage of HVDC transmission is that different power systems can be linked. Any generator linked to an AC grid must exactly match the frequency and phase—they have to be 'synchronous'. This limits the size of a grid. However, unsynchronised grids can be linked by HVDC. The invertor which connects the DC to the new AC system synchronises its output to the new grid. This is indeed how the Victorian and Tasmanian power systems have been linked together by the 400 kV Basslink cable, which consists of one high-voltage undersea cable about 12 cm in diameter and a 9 cm return cable in the same trench. It can transfer over 500 MW of power in either direction between the two states.

It is often said that sustainable energy systems such as wind and solar cannot supply 'base load' electric power. But with the possibility of linking distant systems, for example geothermal from outback Western Australia, wind from the south-west and from Geraldton and solar from Perth rooftops, the need for large baseload stations is reduced considerably.

Safety - RCD Device

Physics in action — Electrical safety

In the last few decades the use of electrical appliances has boomed, while deaths from electrocution have dropped considerably. One reason for this is the use of earth leakage detectors, or *residual current devices* (RCDs). Any appliance used around the home is connected between the active (±340 V) and neutral (0 V) terminals. Normally, an equal amount of current flows in both conductors. If a fault, or carelessness, allows some current to flow to earth through a person, the active and neutral currents are no longer equal. The RCD is designed to detect this and instantaneously shut off the circuit.

The principle is simple: if the currents in each wire are equal but in opposite directions, their magnetic fields will cancel each other. If the currents are not equal, there will be a net magnetic field. In Figure 4.28, the active and neutral conductors pass through a toroidal core. If the currents are equal, no net field is created. If the currents are not equal, the resultant field will induce a current in the pick-up winding. This current activates a solenoid, which switches off the main supply.

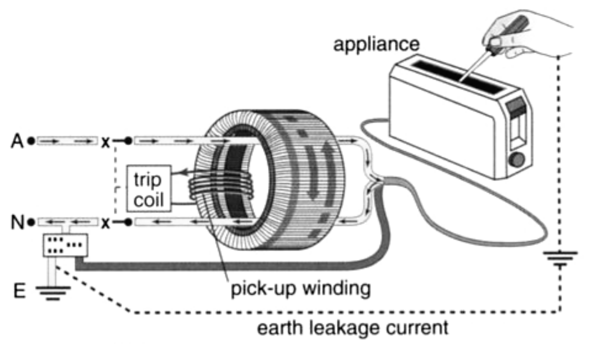


Figure 4.28

Residual current devices prevent fatal shocks by shutting off the current within about 0.03 s.

An RCD cannot eliminate all risk. If you put one hand on a live terminal in the appliance and the other on a neutral wire, the RCD will do nothing because the current going through you is returning through the neutral. For this reason, it is very good practice never to put two hands near a suspect electrical appliance. It is even better practice to disconnect it and take it to an expert!